

A Coupled Mathematical Framework for Twelve-System Neurochemical Dynamics in Cognitive Modeling

Abstract

Computational models of neuromodulation have characterized the roles of individual neurotransmitter systems and their pairwise interactions in cognition and decision-making, but typically operate on single or paired systems, treat receptor subtype dynamics as fixed background parameters, and model brain oscillations separately from underlying neurochemistry rather than coupled to it. We present a coupled mathematical framework for twelve neuromodulator systems that integrates: (i) mean-reverting stochastic kinetics for tonic and phasic concentration components, using an Ornstein–Uhlenbeck drift with a Cox–Ingersoll–Ross-style positivity-preserving square-root diffusion adapted from quantitative finance; (ii) Michaelis–Menten receptor binding across 30 subtypes with a multi-factor effective-signaling proxy that generalizes the operational model of agonism; (iii) a regulatory state machine over four receptor functional states drawn from the canonical GPCR signaling literature; and (iv) an oscillatory closure that derives six-band amplitudes and three cross-frequency coupling envelopes from the neurotransmitter state and back-modulates the kinetic parameters of the lower layers. The coupled system produces multi-timescale dynamics not available from any single layer: three-mechanism habituation under repeated drive, multi-channel cascade attenuation of sustained stimulation, spontaneous emergence of brain-band signatures from neurotransmitter mixtures without explicit state labels, and a contrastive cross-frequency coupling signature across functional regimes. Worked examples demonstrate each behavior with concrete parameter configurations and predict specific empirical consequences of receptor and transporter manipulations including chronic D2 antagonism, 5-HT_{2A} blockade, selective reuptake inhibition, and chronic stress. The framework is hypothesis-generating rather than empirically fit: structural choices and order-of-magnitude parameter ranges are grounded in pharmacology and electrophysiology literature, while specific numerical values are consistency-calibrated and open to refinement. By unifying neurochemical kinetics, receptor pharmacodynamics, and oscillatory dynamics in a single coupled system, the framework provides a substrate for computational work on how neuromodulatory state shapes cognition.

1. Introduction

Cognitive modeling has historically developed within two largely separate traditions. The first, exemplified by symbolic cognitive architectures such as SOAR, ACT-R, and OpenCog, models cognition at the level of computational *function*: representations, rules, problem-solving procedures, learning mechanisms. The second, comprising the bulk of computational neuroscience, models cognition at the level of neural *substrate*: specific anatomical subsystems (basal ganglia, hippocampus, cerebellum), specific neural populations and their dynamics, specific receptor-ligand pharmacodynamics. Neither tradition has provided a satisfactory account of how *neurochemistry* shapes cognition at the system level — how the time-evolving concentrations and pharmacodynamic states of twelve or more neuromodulator systems collectively bias, gate, and reshape cognitive processes across timescales from milliseconds to days. Symbolic architectures abstract over this layer; substrate-level models address slices of it but rarely the whole. Yet the role of neuromodulatory chemistry in cognition is well-established empirically: dopaminergic dynamics shape reward learning, serotonergic state shapes affective valence and behavioral flexibility, noradrenergic state indexes arousal and adaptive gain, cholinergic state modulates learning and attention, and the oscillatory consequences of neuromodulatory state shape the temporal binding and information-routing dynamics of cortex. A model of cognition that does not engage this layer risks losing an important source of explanatory structure.

Within computational neuroscience, a productive research lineage has developed quantitative models of individual neuromodulators and their interactions. The dopamine-as-reward-prediction-error account that originates with Schultz et al. (1997) and is formalized within temporal-difference reinforcement learning (Sutton & Barto, 2018) has been the dominant framework for two decades, and has been extended to D1-D2 dynamics in the basal ganglia (Frank, Seeberger, & O'Reilly, 2004; Frank, 2005), to dopamine-serotonin interactions in motivational and cognitive control (Cools, 2008; Boureau & Dayan, 2011), to serotonergic dynamics in affective and aversive processing (Dayan & Huys, 2009), and to noradrenergic adaptive-gain dynamics (Aston-Jones & Cohen, 2005). The most ambitious integrative theory within this lineage is Doya's (2002) proposal that four neuromodulators (dopamine, serotonin, noradrenaline, acetylcholine) jointly index distinct learning parameters within a unified metalearning framework. This lineage has been productive and we draw on it extensively; but it has three structural limitations. First, most work focuses on single neuromodulators or paired interactions rather than the full multi-system coupling. Second, receptor subtype dynamics (desensitization, internalization, upregulation, subtype switching) are typically treated as fixed background parameters rather than as evolving variables that participate in the cognitive dynamics. Third, oscillatory dynamics, despite their established role in cortical information processing, are typically separated from neuromodulator modeling rather than coupled to it.

This paper presents a coupled mathematical framework for twelve neuromodulator systems that addresses these three limitations simultaneously. The framework

integrates: mean-reverting stochastic kinetics for tonic and phasic concentration components in each of the twelve systems, combining a standard Ornstein–Uhlenbeck drift (Uhlenbeck & Ornstein, 1930) with a Cox–Ingersoll–Ross-style positivity-preserving square-root diffusion adapted from quantitative finance (Cox, Ingersoll, & Ross, 1985), integrated by the Euler–Maruyama scheme (Kloeden & Platen, 1992); a pharmacodynamic receptor layer for 30 receptor subtypes, with Michaelis–Menten binding (Rang, Ritter, Flower, & Henderson, 2019) and a multi-factor effective-signaling proxy that generalizes the operational model of agonism (Black & Leff, 1983); a regulatory state machine over four functional states (active, desensitized, internalized, upregulated) drawn from the canonical GPCR signaling literature (Pierce, Premont, & Lefkowitz, 2002; Gainetdinov et al., 2004); and an oscillatory closure layer that derives six-band amplitudes from the underlying neurochemical state and back-modulates kinetic parameters across the framework’s other layers. The framework’s defining property is that these four layers are *coupled*: the closure rules of the oscillatory layer back-modulate the kinetic parameters of the concentration layer, which co-evolve with the receptor regulatory state and the binding dynamics, producing emergent multi-timescale dynamics that none of the layers can produce in isolation.

The framework’s contribution is not the individual mathematical elements (each a standard tool within its respective tradition) but their simultaneous integration into a single coupled dynamical system that can serve as a substrate for cognitive modeling. We demonstrate through worked examples in Section 4 that this integration produces qualitatively novel behaviors: a three-mechanism account of habituation in which fatigue, adaptive thresholding, and receptor desensitization combine on distinct timescales; a multi-channel cascade attenuation that turns sustained drive into progressively muted downstream signaling without explicit gain-control variables; the spontaneous emergence of band amplitude signatures (resting-state alpha-theta, NREM-analog delta-with-pulsing-sigma) from neurotransmitter mixtures without explicit brain-state labels; and a contrastive cross-frequency coupling signature across functional regimes — a structural prediction, about the sign and segregation of the response rather than its precise magnitude, that the framework offers to empirical work. The framework was developed as the neurochemical substrate of a larger cognitive architecture, but stands on its own as a contribution to computational neuroscience; two companion papers develop the stochastic receptor-state dynamics and the underlying SDE–CTMC methodology in that context.

We are deliberate about what the framework does and does not claim. The framework is *hypothesis-generating* — its structural choices and order-of-magnitude parameter ranges are grounded in the pharmacology and electrophysiology literature, but specific numerical values for closure weights, modulation coefficients, and regulatory thresholds are consistency-calibrated rather than empirically fit. Predictions of Section 4 should be read as predictions of the *shape* of the framework’s response under specific conditions, not as precise empirical fits. The framework’s intended use is as a substrate on which cognitive modeling work can be built — a foundation that other computational

neuroscience researchers can adopt and extend rather than a final-form model of any specific brain region or behavior.

The remainder of this paper proceeds as follows. Section 2 places the framework in the context of prior computational neuroscience work on neuromodulators, prior applications of stochastic differential equations to biological modeling, prior receptor-kinetic models, and prior work on brain oscillations and cross-frequency coupling. Section 3 specifies the framework mathematically across its four coupled layers (3.2 neurotransmitter kinetics, 3.3 receptor binding and effective signaling, 3.4 receptor regulatory state machine, 3.5 oscillatory closure), with §3.1 providing a state-space overview and §3.6 returning to the integrated coupled system. Section 4 demonstrates the framework's behavior across six worked examples spanning steady-state behavior, phasic burst dynamics, receptor adaptation cascades, oscillatory emergence, cross-frequency coupling under task-relevant states, and predictions for specific receptor and neurotransmitter manipulations. Section 5 discusses the framework's relationship to prior work and the methodological implications of its mathematical machinery. Section 6 identifies limitations of the present formulation and directions for future work, both empirical and theoretical.

2. Background and related work

The framework presented in this paper draws on four distinct research traditions: the computational modeling of neuromodulators in decision-making and cognition; the application of stochastic differential equations to biological dynamics; the kinetic modeling of receptor function via continuous-time state representations; and the analysis of brain oscillations and cross-frequency coupling. We review each in turn, situating the framework's contribution within these lineages.

2.1 Computational modeling of neuromodulators

Computational neuroscience has produced a substantial body of work on the role of specific neuromodulators in cognition and decision-making over the past three decades. The dominant thread within this body of work is the dopamine-as-reward-prediction-error tradition, which originates with the seminal characterization of midbrain dopaminergic neuron activity as encoding a temporal-difference prediction error (Schultz et al., 1997) and which has been formalized within reinforcement learning frameworks (Sutton & Barto, 2018; Niv, 2009). Parallel work has developed the role of D1 and D2 receptor subtypes in basal-ganglia learning, distinguishing the contributions of approach and avoidance signaling within the same anatomical structure (Frank, Seeberger, & O'Reilly, 2004; Frank, 2005). This dopaminergic line of work has expanded into pairwise interactions: dopamine-serotonin antagonism in motivational and cognitive control (Cools, 2008; Boureau & Dayan, 2011), the role of serotonin in affective and aversive processing (Dayan & Huys, 2009),

and the role of norepinephrine in adaptive gain and exploration-exploitation arbitration (Aston-Jones & Cohen, 2005; Daw, Niv, & Dayan, 2005).

A small but important strand of work has attempted to integrate multiple neuromodulators into a single theoretical framework. The most direct precedent for the present framework is Doya (2002), which proposed that four neuromodulators (dopamine, serotonin, noradrenaline, and acetylcholine) together index four distinct learning parameters (value-of-reward, exploration-exploitation balance, prediction-error salience, and attentional gain). Doya's metalearning theory is the closest available antecedent to the multi-system coupling principle the present framework formalizes, and we acknowledge it as the conceptual ancestor of our approach. The present framework extends Doya's four-NT theory along three dimensions simultaneously: from four to twelve neuromodulator systems, from a single learning-parameter mapping to a full pharmacodynamic substrate with receptor regulatory dynamics, and from an implicit oscillatory context to an explicit oscillatory closure layer.

A complementary strand of work has applied computational modeling to clinical phenotypes through the lens of computational psychiatry, examining how altered neuromodulatory dynamics might underlie specific cognitive or affective dysfunctions (Dayan & Huys, 2009). This work has produced predictive models of the effects of pharmacological interventions on cognition, but typically operates at the level of single-system or paired-system models. The present framework's contribution to this strand is to provide a substrate on which more specialized clinical-prediction models can be built without each model needing to re-derive the underlying neurochemical dynamics independently.

2.2 Stochastic differential equations in biological modeling

The mathematical machinery the framework uses (the Ornstein–Uhlenbeck process, the square-root multiplicative diffusion, the Euler–Maruyama integration scheme) has a development history that spans physics, neuroscience, finance, and mathematical biology. The Ornstein–Uhlenbeck process was originally introduced to describe the velocity of a Brownian particle subject to viscous friction (Uhlenbeck & Ornstein, 1930) and is already native to computational neuroscience: it is the standard model of subthreshold membrane-potential fluctuations and the engine of the drift-diffusion family of decision models. We therefore do not present the OU drift as a novel import; it is the natural and well-established choice for mean-reverting neural quantities, and we adopt it as such. The square-root diffusion structure, by contrast, was developed within quantitative finance as a model for nonnegative interest-rate dynamics (Cox, Ingersoll, & Ross, 1985), where the diffusion form was chosen specifically to enforce positivity of the integrated process. This positivity-preserving multiplicative-noise structure is comparatively underused in computational neuroscience, where additive noise is the more common default, and it is this specific element, not the OU drift, that the framework adapts from the finance literature. The numerical-analysis foundations for integrating SDEs with these structures are due to Kloeden and Platen (1992) and remain the canonical reference for Euler–Maruyama and related schemes.

Within neuroscience, SDE methods have been applied predominantly to two contexts: ion-channel noise modeling at the membrane level, and diffusion-approximation analyses of spiking-network mean-field dynamics. The square-root (Feller) diffusion specifically is also not unprecedented in neuroscience: it was proposed decades ago as a model of subthreshold membrane potential between spikes (Giorno et al., 1988), where it was argued to be more realistic than the Gaussian Ornstein–Uhlenbeck model precisely because it is bounded below. Our use of it is therefore distinguished not by the process but by the *level* of description: prior square-root models describe the membrane voltage of a single neuron, whereas the present framework applies the structure to the bulk concentration of a neurotransmitter pool, a quantity at a different level of biological organization with its own positivity and rate-scaled-variability rationale. The use of CIR-style positivity-preserving multiplicative noise at this concentration level is, to our knowledge, novel; we make that level-specific claim rather than claiming the process is new to neuroscience, which it is not.

2.3 Receptor kinetics and continuous-time state models

The treatment of receptor function as a kinetic process governed by transitions between discrete functional states originates in the analysis of ligand-gated ion channels (Colquhoun & Hawkes, 1977, 1981), where the stochastic open-closed gating dynamics of single channels were first formalized as continuous-time Markov chains. This framework has remained the canonical mathematical structure for receptor kinetics for nearly five decades. Subsequent developments extended the CTMC approach to synaptic receptors with more elaborate state machinery (Destexhe, Mainen, & Sejnowski, 1994) and to the explicit modeling of receptor regulatory state changes (desensitization, internalization, recycling, upregulation) within the GPCR signaling literature (Pierce, Premont, & Lefkowitz, 2002; Gainetdinov, Premont, Bohn, Lefkowitz, & Caron, 2004; Hanyaloglu & von Zastrow, 2008).

A parallel pharmacological lineage developed the operational model of agonism (Black & Leff, 1983), which decomposes pharmacological response into the product of occupancy and intrinsic efficacy and remains the canonical framework for relating ligand binding to downstream signaling. The textbook treatment of receptor pharmacology (Rang, Ritter, Flower, & Henderson, 2019) consolidates both the kinetic and the operational lineages within a unified pharmacological framework.

The present framework's contribution to this lineage is the explicit integration of a CTMC-style functional state machine with a multi-factor effective-signaling proxy (eq. 22) that decomposes pharmacological efficacy into its mechanistic sub-components: density, sensitivity, G-protein coupling, and state. This generalizes the operational model by factorizing efficacy and by coupling the factorization to regulatory dynamics that update the sub-components as functions of binding history. The present paper uses a deterministic, threshold-triggered form of this state machine, which we argue (§3.4) is the appropriate level of description for population-level regulatory dynamics; a companion paper develops a fully stochastic rate-kernel generalization as an extension.

2.4 Oscillatory coupling and cross-frequency dynamics

The fourth research tradition the framework draws on is the analysis of brain oscillations and cross-frequency coupling. The systematic study of brain rhythms across mammalian cortex (their characteristic frequencies, their relationships to behavioral state, and their interactions across spatial and frequency scales) is the subject of a substantial neuroscience literature reviewed comprehensively in Buzsáki (2006). Subsequent work has developed band-specific functional theories: theta as a substrate for episodic encoding and temporal sequencing (Buzsáki, 2002; Lisman & Grace, 2005; Vertes, 2005), alpha as a substrate for inhibitory gating (Klimesch, 2012), beta as a substrate for attentional control and the maintenance of task sets (Engel & Fries, 2010), gamma as a substrate for binding and integration (Bartos, Vida, & Jonas, 2007; Buzsáki & Wang, 2012), delta and the sigma spindle band as substrates for slow-wave consolidation (Steriade, McCormick, & Sejnowski, 1993; Steriade, 2006; Mölle & Born, 2011).

Cross-frequency coupling has emerged as a distinctive area of theoretical interest within this literature. The theta-gamma coupling that has been most extensively characterized, both empirically and as a candidate neural code for associative encoding (Lisman & Jensen, 2013), is the prototypical case, but a broader theoretical literature has developed around the functional roles and mechanistic origins of CFC across multiple band pairs (Canolty & Knight, 2010; Hyafil, Giraud, Fontolan, & Gutkin, 2015). Methodological work has standardized the measurement of phase-amplitude coupling and related quantities (Tort, Komorowski, Eichenbaum, & Kopell, 2010).

Generative models of band activity from a biophysical substrate do exist — most prominently the neural-mass and dynamic-causal-modeling lineage (Jansen & Rit, 1995; David & Friston, 2003; Moran et al., 2007), which generates band spectra from simulated population dynamics and has incorporated neuromodulatory effects (Moran et al., 2013). What this literature has not typically provided is a *direct, closed-form* mapping from neuromodulator concentrations to band amplitudes that bypasses population-dynamics simulation altogether. In the neural-mass tradition the rhythm is an emergent property of interacting excitatory–inhibitory populations, with chemistry entering as parameters; the question of which band a given neuromodulator profile produces is answered only by simulating the dynamics. The framework’s closure rules of §3.5.2 instead provide an explicit algebraic per-band derivation directly from the twelve-NT concentration vector, with each entry biologically motivated and each resulting structural prediction (Sections 4.4–4.5) — the sign and relative ordering of band responses, not their hand-tuned magnitudes — open to empirical test. The contribution relative to prior oscillation modeling is this tractable directness and its couplability to the framework’s other layers, not the existence of a chemistry-to-rhythm relationship, which the neural-mass lineage already established.

2.5 Summary

The framework integrates contributions from these four research traditions: it extends the multi-neuromodulator integrative theory of Doya (2002) to twelve systems and adds receptor and oscillatory layers; it adapts the CIR-style positivity-preserving square-root diffusion from quantitative finance to a neurochemical substrate, combined with the neuroscience-native Ornstein–Uhlenbeck drift; it generalizes the CTMC receptor-kinetic and operational-pharmacology lineages with a multi-factor effective-signaling proxy and regulatory dynamics; and it provides a *direct algebraic closure* from neuromodulator concentrations to oscillatory band amplitudes — distinct from the simulation-based generative models of the neural-mass and dynamic-causal-modeling lineage in being closed-form and directly couplable to the concentration and receptor layers, rather than emergent from population dynamics. Each component of the framework is independently defensible within its parent tradition; the framework’s distinctive value is the simultaneous integration of all four into a single coupled dynamical system. The remainder of this paper develops the mathematical specification (Section 3), the resulting model behavior (Section 4), and the conceptual and empirical implications (Sections 5 and 6).

3. Model

3.1 Overview and state variables

The framework is a coupled stochastic dynamical system organized into four interacting layers. Before specifying each layer in detail, we enumerate the state space and the relationships between the layers, both to orient the reader and to make the framework’s overall dimensionality explicit.

The twelve neurotransmitter systems. The framework models twelve neurotransmitter and neuromodulator systems, selected to span the major neuromodulatory chemistry of mammalian central nervous system signaling: dopamine (DA), serotonin (5-HT), norepinephrine (NE), acetylcholine (ACh), gamma-aminobutyric acid (GABA), glutamate (GLU), oxytocin (OXT), the mu-opioid system (MOR), the endocannabinoid system (CB1), histamine (HIST), corticotropin-releasing hormone (CRH), and cortisol (CORT). Each system is represented by four scalar state variables: the tonic concentration, the phasic concentration, the fatigue variable, and the transporter-efficiency variable. The twelve systems are coupled through the receptor and oscillatory layers below rather than through direct concentration-to-concentration terms; we discuss this design choice and its principal simplification (the omission of direct presynaptic heteroreceptor coupling) below.

The receptor population. The framework instantiates 30 receptor subtypes spanning eleven of the twelve transmitter systems, with subtype counts ranging from 1 to 5 per transmitter depending on the biological complexity of the system (cortisol is modeled as concentration-only, signaling through direct modulation with no discrete receptor

subtype; a complete enumeration is provided in Appendix B). Each receptor subtype carries six per-receptor state variables: density, sensitivity, G-protein coupling efficacy, discrete functional state, exposure trace, and the time-in-state counter used by the regulatory transitions. The receptor population therefore contributes scalar state variables (including the categorical encoded as an integer index).

The oscillatory layer. Six band amplitudes (delta, theta, alpha, beta, gamma, sigma) and three cross-frequency coupling envelopes, complete the framework's state representation. The amplitudes are derived quantities of the concentration vector through the closure rules of §3.5.2 and do not constitute independent dynamical variables; we track them as state variables in the implementation for efficiency and instrumentation but they are computed at each timestep rather than evolved by independent equations. Phase variables for each band are tracked but do not enter the present formulation's modulation rules (see §6 for discussion of phase-resolved extensions).

Total state dimensionality. Aggregating across the three layers, the framework's instantaneous state vector contains approximately scalar variables when fully instantiated, plus the categorical state for each receptor subtype. This is large compared to many computational neuroscience models but small relative to spiking-network simulations, and the dimensionality is well-suited to the framework's intended use as a substrate for cognitive modeling.

The equations of motion. The framework's dynamics are specified by a coupled system of stochastic and deterministic equations: a mass-balance kinetic equation per transmitter (eqs. 1, 4, 5), with three loss channels (eq. 6), multiplicative square-root noise (eq. 7), and fatigue dynamics (eq. 15); a binding equation per receptor (eqs. 19–20) with multiplicative effective-signaling output (eq. 22) and exposure trace (eq. 23); a regulatory state machine per receptor with six directed transitions (eqs. 24–29); a band-closure rule per band (eq. 32) and three cross-frequency products (eqs. 33–35); and six modulation rules (eqs. 36–41) that close the loop back to the lower layers. The full coupled system therefore comprises on the order of 80 first-order coupled equations under default instantiation, evolved by an adaptive-step Euler–Maruyama integrator (eq. 18) with seeded pseudorandom number generation for deterministic reproducibility.

Scope of the modeled dynamics, and an important coarsening. Two features of the framework's scope must be made explicit at the outset, because they bound what the framework does and does not claim to capture. First, the twelve systems span a wide range of signaling biology, and they are *not* dynamically equivalent. Glutamate and GABA signal predominantly through fast ionotropic receptors (AMPA, NMDA, kainate; GABA-A) on millisecond timescales, with only a minority metabotropic component (mGluR, GABA-B); cortisol and CRH act partly through nuclear steroid receptors with genomic effects unfolding over minutes to hours to days; and the monoamines and neuropeptides signal predominantly through the G-protein-coupled receptors whose regulatory cycle (desensitization, internalization, recycling, upregulation) the state machine of §3.4 is designed to model. The framework tags each of the 30 receptor

subtypes as ionotropic or metabotropic, but in the present formulation it applies the same slow regulatory state machine uniformly across both classes. This is a deliberate coarsening, and its justification is that the framework models the *slow neuromodulatory envelope* of each system (the timescale on which concentrations bias, gate, and reshape downstream cognition) rather than the fast synaptic transmission events that ionotropic receptors mediate. The framework is therefore not a model of fast excitatory or inhibitory postsynaptic dynamics, and its glutamate and GABA components should be read as capturing their slow modulatory and homeostatic roles (including the genuinely metabotropic mGluR and GABA-B signaling and the slower regulatory adaptation of the ionotropic populations) rather than their millisecond channel kinetics. Likewise, the cortisol and CRH components model a slow concentration-and-sensitivity envelope as a proxy for the genomic and non-genomic stress-axis effects, not the full receptor biology of nuclear steroid signaling. We regard differential dynamics for ionotropic and steroid receptors (a fast-channel model for the former, a genomic-delay model for the latter) as identified future work (§6), and we note that the existing ionotropic/metabotropic tagging in the implementation provides the natural hook for it. Readers should interpret all predictions involving the glutamate, GABA, cortisol, and CRH systems as predictions about their slow modulatory contribution, not their fast or genomic signaling per se.

Second, the framework models inter-system coupling exclusively through shared receptor and oscillatory state, with no direct concentration-to-concentration kinetic term — there is no equation in which the concentration of one transmitter directly enters the drift of another. This is a simplification: presynaptic heteroreceptor regulation, in which one transmitter directly modulates the release of another, is well documented biologically and is *not* represented at the concentration level in the present framework. The justification is again one of scope and parsimony: much of the functional consequence of heteroreceptor cross-talk is captured indirectly, because a transmitter that modulates another’s receptors or shifts the shared oscillatory state thereby influences the other’s effective signaling. But this is an approximation, not an identity, and direct presynaptic NT–NT coupling is a candidate extension we flag explicitly rather than one the framework currently captures.

Notation. A few Greek symbols carry more than one meaning across the framework’s four layers; they are always disambiguated by subscript, superscript, or argument, and we collect the reusable ones here for reference. Band amplitudes are *always* written with an explicit band subscript ω , so the oscillation-band reading of any symbol below is never the bare letter.

Symbol	Distinct uses	Disambiguated by
	OU mean-reversion <i>rate</i> (λ); regulatory and release <i>thresholds</i> ($\theta_1, \theta_2, \dots$); the <i>theta band</i> (θ)	subscript; band only as ω

Symbol	Distinct uses	Disambiguated by
	diffusion/noise <i>term</i> () and its tabulated <i>amplitude</i> (,); receptor <i>sensitivity</i> state (); OU stationary <i>standard deviation</i> ; the <i>sigma band</i> ()	superscript = sensitivity; band only as
	G-protein <i>coupling efficacy</i> (); coupling/fatigue <i>rate</i> constants (, ,); the <i>gamma band</i> ()	superscript/subscript; band only as
	temporal-difference/RPE <i>signal</i> (); rate constants (,); the <i>delta band</i> ()	argument = RPE; band only as
	per-NT noise <i>coefficient</i> (, eq. 7); oscillatory modulation <i>coefficients</i> (, , ,); the <i>alpha band</i> ()	subscript; band only as
	fatigue→reversion coupling coefficient (eq. 16); i.i.d. standard-normal Euler–Maruyama draw (eq. 18)	argument = noise draw

Section structure. The remainder of this section specifies each layer in turn: §3.2 develops the kinetic dynamics, §3.3 the receptor binding and effective-signaling proxy, §3.4 the regulatory state machine, and §3.5 the oscillatory closure. §3.6 returns to the integrated coupled system. The development is bottom-up (from chemistry to signaling to regulation to rhythms) to facilitate sequential reading; readers familiar with any individual layer may skip to the integration in §3.6 without loss of the framework’s overall structure.

3.2 Neurotransmitter kinetics

3.2.1 Mass-balance evolution

For each of the twelve modeled neurotransmitter systems, indexed by i , the synaptic concentration x_i evolves according to a mass-balance kinetic equation comprising one input (release) and three clearance channels (transporter reuptake, enzymatic degradation, and diffusion/spillover):

This decomposition mirrors the established biological mechanisms of synaptic neurotransmitter clearance (Kandel et al., 2013), separating channels that operate on

distinct timescales and through distinct molecular machinery: transporter-mediated reuptake (e.g., DAT for dopamine, SERT for serotonin, NET for norepinephrine) on the order of hundreds of milliseconds, enzymatic degradation (e.g., COMT, MAO-A/B, AChE) on a comparable timescale, and passive diffusion or volume transmission on slower scales. Treating these channels independently allows the framework to express transmitter-specific clearance profiles via three separate kinetic coefficients rather than a single lumped decay rate, which is necessary to capture the well-documented differences between, for example, the predominantly reuptake-driven clearance of dopamine and the predominantly enzymatic clearance of acetylcholine (Stahl, 2013).

The concentration variable is further decomposed into tonic and phasic components, reflecting the canonical distinction between baseline signaling and transient burst events (Grace, 1991):

Each component is integrated as a separate state variable with its own reversion target and noise amplitude, but with shared loss kinetics. This separation is functionally motivated: tonic dopamine encodes long-horizon valuation while phasic bursts encode reward prediction errors (Schultz et al., 1997; Niv, 2009); analogous tonic-phasic distinctions are documented for noradrenaline (Aston-Jones & Cohen, 2005), acetylcholine (Hasselmo, 2006), and other neuromodulatory systems. By representing both components explicitly, the framework preserves the ability of downstream receptor populations to respond differentially to baseline versus burst signaling (Section 3.3).

3.2.2 Stochastic differential form

We treat eq. (1) as a stochastic differential equation, decomposing the right-hand side into a deterministic drift term and a stochastic diffusion term :

where ΔW is the increment of a standard Wiener process driving the noise.

The drift term combines a mean-reverting Ornstein–Uhlenbeck contribution (Uhlenbeck & Ornstein, 1930) with the multi-channel loss term. For the tonic component, reversion is toward the homeostatic baseline :

For the phasic component, reversion is toward zero rather than the baseline, reflecting its interpretation as a transient deviation from tonic signaling:

The total loss term aggregates the three clearance channels of eq. (1):

where the extracellular reference concentration defaults to zero in the single-compartment formulation used throughout this paper, and η is a fatigue-modulated transporter efficiency defined in Section 3.2.4.

The diffusion term uses a multiplicative square-root noise model:

with σ a per-transmitter noise amplitude coefficient (further modulated by oscillatory state; see Section 3.5). The square-root form is principled rather than ad hoc: it is the same diffusion structure used in the Cox–Ingersoll–Ross interest-rate model (Cox, Ingersoll, & Ross, 1985), where it ensures positivity of the integrated process. Biologically, it is motivated by the Poisson-like character of vesicular release variability — the standard deviation of release events scales as the square root of the mean release rate. Numerically, eq. (7) has the additional virtue of vanishing at the lower boundary: as $x \rightarrow 0$, the stochastic contribution vanishes, preventing the integrator from producing negative concentrations purely through noise excursions. We return to this property in Section 3.2.5 when discussing boundary handling.

The lineage of these two forms (the OU drift native to computational neuroscience, the square-root diffusion adapted from quantitative finance) is discussed in §2.2. The short version: neurotransmitter concentrations share exactly the structural properties the CIR form was designed for (bounded below by zero, mean-reverting, and subject to release variability whose magnitude scales with the rate) which is what motivates its use here (see also §5.4 for the methodological framing).

3.2.3 Release drives

The release term in eqs. (4–5) is computed per-transmitter from a module specification that maps upstream cognitive signals to a release rate. Let $\mathbf{u}$ denote the vector of available upstream signals (novelty, reward prediction error, effort demand, emotion intensities, and so on); the raw release for transmitter i is the threshold-rectified weighted sum:

Concrete drive functions for the principal release signals are:

where δ is a temporal-difference prediction error (Sutton & Barto, 2018), with biological grounding in midbrain dopamine signaling (Schultz et al., 1997); s is a stimulus-novelty signal; and t is a task-demand signal. Emotion-derived drives follow a per-transmitter recipe table (see Supplementary §S2). The signal-to-transmitter weight matrix $\mathbf{W}$ encodes the framework’s assumptions about which cognitive signals drive which neurochemical

responses, and is the principal point of contact between the substrate layer and any downstream cognitive architecture that consumes it.

The raw release is then subject to two post-processing operations before entering eqs. (4–5). First, **fatigue gating** suppresses release under sustained high activity:

Correction (implementation discrepancy): the shipped implementation does not evaluate the expression above directly. It normalizes the suppression term by $(1-F^*)$ and floors the resulting multiplier at zero, which is the form actually used to generate this paper's worked-example figures:

Under this corrected form, full suppression ($R_i^{\text{fat}} = 0$) is reached exactly at $F_i = 1$ — the documented maximum of the fatigue variable — rather than at $F_i = 1.7$, and gated release cannot go negative for any F_i , including out-of-domain values. The equation above should be read as superseded by this corrected form; the surrounding threshold description in the following paragraph continues to hold.

with default and threshold , so that fatigue below the threshold has no effect and above it scales the release linearly toward zero. Second, **oscillatory gating** scales release multiplicatively by band amplitudes (Section 3.5):

with the most prominent default coupling being , reflecting the empirically documented role of gamma-band activity in gating phasic dopaminergic release. Phasic burst amplitudes additionally saturate exponentially in the drive strength:

Together, eqs. (8–14) specify the input term to eqs. (4–5).

3.2.4 Fatigue dynamics

Each transmitter system carries a fatigue variable that accumulates with activity and decays passively:

with default rates and , giving accumulation over tens of timestep-equivalents of sustained activity and passive recovery over hundreds. The update is clipped to the stated domain after each step, $F_i(t+\Delta t) = \text{clip}_{[0,1]}(F_i(t) + dF_i)$; the clip is load-bearing rather than defensive, since the unclipped steady state under constant drive is $F_{ss} = (\epsilon_i/\delta_{F,i}) \cdot C_i = 10 \cdot C_i$, so without it any sustained concentration above ≈ 0.1 — well within the operating range this paper's own worked examples use — would drive fatigue past its documented upper bound. The fatigue variable enters the dynamics through two channels. First, it modulates the effective mean-reversion rate:

with default τ , slowing reversion under high-activity conditions. Second, it modulates transporter efficiency η , which appears in the loss term eq. (6):

with default τ , attenuating reuptake under fatigue. The combination of eqs. (15–17) produces system-level homeostatic behavior: sustained drive accumulates fatigue, which both slows return to baseline and attenuates clearance, allowing transient elevation of concentration but eventually saturating release through eq. (12).

3.2.5 Numerical integration

The coupled SDE system, eq. (3) with drift and diffusion specified by eqs. (4–7), is integrated using the Euler–Maruyama scheme (Kloeden & Platen, 1992):

with ξ drawn i.i.d. per timestep and per transmitter. The default step size is Δt (in normalized units), chosen to satisfy the standard Euler–Maruyama stability condition over the operating range of the model.

The boundary condition C_{lower} is enforced after each step. Naïve iterative clamping (reflect, check, reflect again) is known to diverge when the unconstrained step would land far outside the domain, because successive reflections can amplify rather than converge. We use instead an modular-folding scheme:

```
shifted      = (C_next - lower) mod (2 * domain_width)
if shifted > domain_width: shifted = 2 * domain_width - shifted
C_bounded    = lower + shifted
```

This maps any real-valued C into the bounded domain in a single arithmetic operation, while preserving the stationary distribution of the unbounded process over the bounded region. The reflecting boundary preserves probability mass that would otherwise be lost to absorption and avoids the variance distortion introduced by hard clipping.

The framework additionally supports adaptive step-sizing via Richardson extrapolation: at each step, a single full step of size Δt is compared against two half-steps of size $\Delta t/2$ traversing the same Brownian path (constructed by splitting the increment as ξ with $\xi/2$). When the discrepancy exceeds a configured tolerance, the step is rejected and retried at $\Delta t/2$; otherwise it is accepted, with Δt adjusted upward toward a maximum bound for the following step. Full numerical details are provided in Appendix D.

All stochastic integration in the framework uses seeded pseudorandom number generation (`numpy.random.default_rng` with explicit seed), ensuring deterministic, byte-reproducible trajectories under identical configuration. The implementation includes a comprehensive suite of unit and integration tests verifying that the code reproduces

the dynamical properties asserted above — fidelity to the equations as specified, not biological validity (see §6.3). See *Code Availability* and Supplementary §S1 for details.

3.3 Receptor binding and effective signaling

The concentration dynamics of Section 3.2 produce the upstream signal that drives downstream computation, but concentrations themselves are not yet signaling: they must bind to receptors, and the bound-receptor population must produce a downstream effect. We model this transduction with three components: (i) a localization-weighted effective concentration that accounts for where a receptor sits relative to the release site, (ii) a Michaelis–Menten saturation function mapping effective concentration to fractional occupancy, and (iii) a multiplicative effective-signaling proxy combining occupancy with receptor density, sensitivity, G-protein coupling efficacy, and functional state. The framework instantiates 30 receptor subtypes spanning eleven of the twelve transmitter systems, each carrying its own copy of this transduction chain with subtype-specific parameters.

3.3.1 Localization and effective concentration

Receptors at different synaptic positions are exposed to different mixtures of the tonic and phasic concentration components defined in eq. (2). Synaptic and perisynaptic receptors are physically close to the release site and respond to both components; extrasynaptic receptors, in contrast, are reached only by neurotransmitter that has escaped the synaptic cleft via diffusion or volume transmission, and therefore respond predominantly to the tonic pool (Fuxe et al., 2010; Agnati et al., 2010). We capture this position-dependence with a per-receptor localization parameter and define the effective concentration sensed by receptor on transmitter as:

The limit cases are interpretable:

- (presynaptic/synaptic): the receptor sees the full — both baseline and burst components contribute.
- (perisynaptic): the receptor sees the full tonic component plus a partial phasic contribution.
- (extrasynaptic): the receptor sees only — phasic bursts are too spatially localized to reach this population.

The biological motivation for this single-parameter localization rule comes directly from the volume-transmission literature. Fuxe and colleagues established that extrasynaptic receptors operate on slower, lower-amplitude tonic signals and are functionally specialized for state-setting and modulation rather than phasic event-coding (Fuxe et al., 2010; Agnati et al., 2010). The framework respects this distinction at the level of the binding equation: extrasynaptic receptors of the same subtype as synaptic ones bind

less ligand during phasic events, not because their affinity differs but because the relevant concentration is structurally different.

3.3.2 Ligand binding and saturation

We model fractional receptor occupancy as a Michaelis–Menten function of the effective concentration:

with the per-subtype dissociation constant (the half-saturation concentration). Eq. (20) is the Hill equation in the non-cooperative limit ($n=1$) and is the standard equilibrium binding model for non-cooperative agonist binding to a single receptor population (Rang et al., 2019). We use K_d throughout the framework as the default; cooperative binding ($n > 1$) is biologically documented for a small number of receptor populations (notably some ligand-gated ion channels) but is not used in the present formulation. The framework is straightforwardly extensible to cooperative binding by replacing eq. (20) with the full Hill form .

The dissociation constant K_d is treated as a per-subtype parameter calibrated from the receptor pharmacology literature. Its baseline values span roughly two orders of magnitude across the 30 modeled subtypes, reflecting the wide dynamic range of biological binding affinities. K_d is not strictly static, however: oscillatory state modulates apparent affinity through allosteric and conformational mechanisms (theta-band modulation of dopaminergic affinity has been particularly well-characterized; see Section 3.5). We defer the full modulation rule to eq. (36) in Section 3.5 and treat K_d as a constant in this section's equations for clarity.

3.3.3 Receptor state and the effective-signaling proxy

Occupancy is necessary but not sufficient for signaling: a bound receptor produces a downstream effect only if it is in a competent functional state, expressed at sufficient density on the cell surface, intrinsically sensitive at the moment of binding, and successfully coupled to its G-protein (or, for ionotropic receptors, its associated ion channel). We capture these factors with four additional per-receptor state variables that together determine the effective signaling output:

- **Density** : the surface expression of receptor R , normalized to $R_{baseline}$ at baseline and transiently boosted above $R_{baseline}$ under upregulation (§3.4.4). Density changes on slow timescales through internalization and recycling (Section 3.4); on faster timescales it is treated as approximately constant.
- **Sensitivity** (normalized to $S_{baseline}$ at baseline, transiently exceeding $S_{baseline}$ under upregulation): a phenomenological compound parameter aggregating phosphorylation state, allosteric modulator occupancy, and other mechanisms that modulate the response per unit occupancy. Sensitivity declines on entry to the desensitized state and recovers when the receptor returns to ACTIVE (Section 3.4).

- **G-protein coupling efficacy** : the efficiency with which an active, sensitive, bound receptor recruits and activates its associated G-protein. Coupling efficacy degrades under sustained high occupancy (a well-documented consequence of GRK-mediated phosphorylation and arrestin recruitment; Pierce, Premont, & Lefkowitz, 2002) and recovers slowly when occupancy drops. The framework imposes a floor to model the residual basal coupling observed in vivo.
- **Functional state** : a discrete categorical variable indicating the receptor's regulatory state. State-dependent signaling efficacy is captured by a gating function :

The gate values reflect canonical receptor regulatory phenomenology: full signaling in the active state, partial signaling under desensitization, near-zero signaling once the receptor has been internalized to endosomal compartments, and supra-baseline signaling in the upregulated state that follows prolonged understimulation. The dynamics that govern transitions between these four states are the subject of Section 3.4 and are not specified here; for the purposes of the binding equation, is treated as the receptor's current state and as the corresponding multiplicative gate.

With these four state variables in hand, we define the **effective signaling proxy** of receptor on transmitter as:

The multiplicative structure reflects the conceptual logic that *every* factor in the chain must be non-zero for the receptor to produce signal: a receptor with high occupancy but zero density signals nothing, as does one with high density and zero coupling efficacy. This generalizes the operational model of agonism (Black & Leff, 1983), which decomposes pharmacological response into the product of occupancy and intrinsic efficacy, by additionally factorizing efficacy into its mechanistic sub-components (density, sensitivity, G-protein coupling, functional state). Under baseline normalization () the proxy is bounded above by , i.e. by in the UPREGULATED state; because the upregulation transition additionally boosts density and sensitivity above their baseline of (§3.4.4), the realized proxy of a fully upregulated, fully bound receptor can reach (§4.3.2). In all cases approaches zero whenever any single factor approaches zero.

The framework treats as the receptor's contribution to downstream computation, but does not itself specify how downstream cognitive systems consume the signal. Different cognitive architectures may aggregate the 30 per-receptor signals with subtype-specific weights, route them to compartmentalized computational modules, or use them as gain parameters on other processes. Eq. (22) defines the substrate; the downstream layer defines its consumption.

3.3.4 Exposure trace

A final auxiliary variable derived from binding is the **exposure trace**, a leaky integral of saturation that captures cumulative ligand exposure on a timescale slower than instantaneous occupancy:

with default time constant (in normalized units, matching the timescale over which receptor regulatory state changes are biologically plausible). The exposure trace is not itself a signaling variable — it does not enter eq. (22) — but it drives the CTMC transitions that determine , as developed in the next section. Conceptually, encodes the history of binding that the receptor’s regulatory machinery uses to decide whether to remain in its current state, downregulate, or upregulate.

The introduction of closes the binding layer: from a transmitter concentration trajectory and a fixed configuration of receptor parameters, we can compute the time-varying effective signaling output and the slower exposure trace that governs receptor regulatory dynamics. Together, the equations of this section provide the bridge from the chemistry of Section 3.2 to the receptor regulatory dynamics of Section 3.4 and the cross-frequency coupling of Section 3.5.

3.4 Receptor functional-state dynamics

The state variables introduced in Section 3.3 (density , sensitivity , G-protein coupling efficacy , and functional state) evolve on slower timescales than the millisecond-scale concentration and binding dynamics. In this section we specify the regulatory dynamics that determine and the coupled adjustments to , , and that accompany transitions between functional states. The framework treats this regulatory machinery as a discrete-state machine over four functional states with threshold-driven transitions on a continuous-time substrate. The state variable is discrete, but the accumulator variables that govern its transitions (cumulative ligand exposure , time-in-state , and instantaneous saturation) evolve continuously between transition events.

We adopt deterministic, threshold-triggered transitions in this formulation, and we treat that as a deliberate modeling choice rather than a placeholder. A deterministic state machine is fully adequate for the framework’s purpose (representing the slow regulatory envelope of receptor populations whose aggregate behavior is well-approximated by threshold crossings of accumulated exposure) and it has the practical virtues of being analytically transparent, computationally cheap, and exactly reproducible. We use continuous-time Markov chain notation to describe it, in line with the standard receptor-kinetics literature (Colquhoun & Hawkes, 1977, 1981); in this notation the present model is the limiting case in which transition rates are step functions of the accumulator variables (zero below threshold, effectively instantaneous above). A fully stochastic generalization, in which transitions follow smooth rate kernels and individual-receptor trajectories fluctuate around the population mean, is a natural extension that the notation already accommodates, and we develop it in a companion

paper; but nothing in the present paper depends on that generalization, and the deterministic model stands on its own as the appropriate level of description for population-level regulatory dynamics.

3.4.1 The state set and its biological interpretation

We model four functional states corresponding to canonical regulatory phenomenology documented across G-protein-coupled receptors (Pierce, Premont, & Lefkowitz, 2002; Gainetdinov, Premont, Bohn, Lefkowitz, & Caron, 2004) and partially generalized to ionotropic receptors (Hanyaloglu & von Zastrow, 2008):

- **ACTIVE** (): the receptor is surface-expressed, properly coupled to its associated G-protein or ion channel, and competent to signal. This is the default state for an unstimulated or moderately stimulated receptor.
- **DESENSITIZED** (): the receptor remains surface-expressed but signals at attenuated efficacy (default). Biologically, this corresponds to GRK-mediated phosphorylation followed by arrestin recruitment that displaces the G-protein (Lefkowitz, 2007). The receptor is still capable of binding ligand but the binding does not produce full downstream signaling.
- **INTERNALIZED** (): the receptor has been removed from the plasma membrane and resides in endosomal compartments. Effective signaling is near-zero (default), and the receptor must be either recycled back to the surface or degraded. The 10% residual reflects the small fraction of internalized receptors that retain partial signaling competence in endosomes (Hanyaloglu & von Zastrow, 2008).
- **UPREGULATED** (): the receptor population has expanded above baseline density via reduced turnover or increased synthesis, typically following prolonged understimulation. Effective signaling is enhanced (default) above the ACTIVE baseline. This state corresponds to the receptor-supersensitivity phenomena observed clinically after sustained antagonist exposure and in withdrawal states.

The choice of four states is deliberately parsimonious. Biological receptor regulation involves a much larger number of intermediate molecular states (multiple phosphorylation patterns, distinct arrestin conformations, multiple endosomal trafficking subpopulations), but the four-state coarsening captures the functional-effect categories that matter for downstream signaling and is the standard granularity used in pharmacological textbook treatments (Rang et al., 2019). Finer-grained state structure can be added without disturbing the framework architecture.

Receptor regulatory state machine

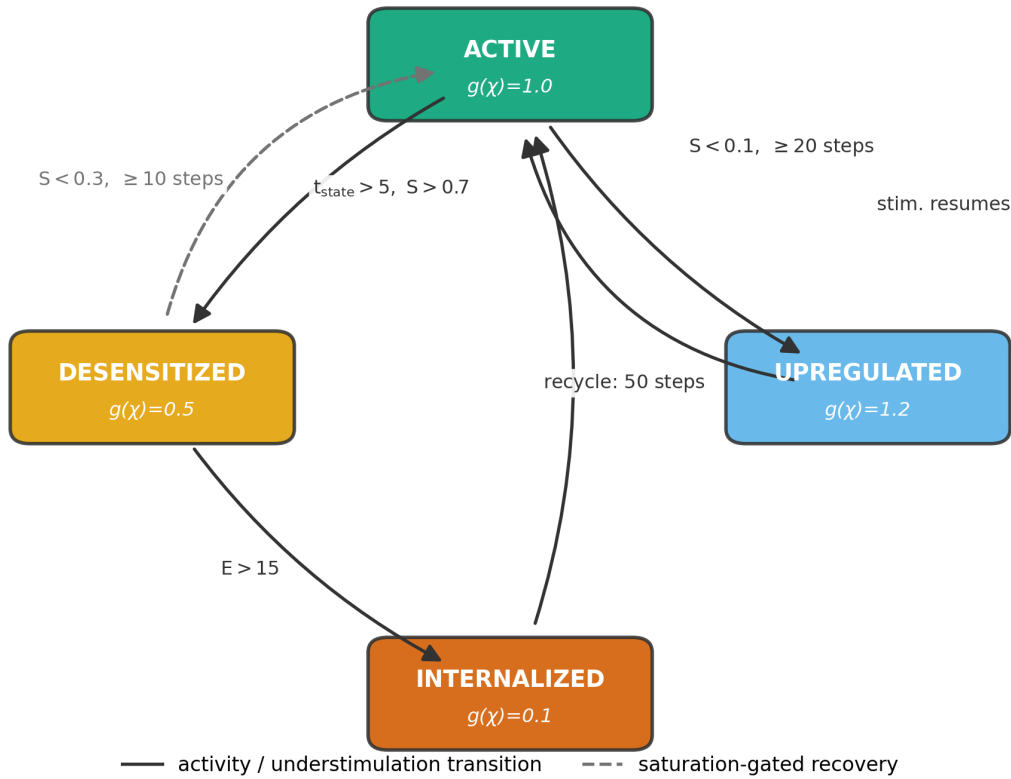


Figure 3.4.1: Receptor regulatory state machine

Figure 3.4.1. Receptor regulatory state machine. Four functional states (ACTIVE, DESENSITIZED, INTERNALIZED, UPREGULATED) and six directed transitions. Solid arrows: activity-driven cycle (ACTIVE $\rightarrow$ DESENSITIZED when $S < 0.3$ and ≥ 10 steps; DESENSITIZED $\rightarrow$ INTERNALIZED when cumulative exposure $E > 15$; INTERNALIZED $\rightarrow$ ACTIVE after 50 recycling steps) and understimulation cycle (ACTIVE $\rightarrow$ UPREGULATED when $S < 0.1$ and ≥ 20 steps; UPREGULATED $\rightarrow$ ACTIVE when stimulation resumes). Dashed arrow: saturation-gated recovery shortcut (DESENSITIZED $\rightarrow$ ACTIVE when $t_{\text{state}} > 5$ and $S > 0.7$ steps). State-gate values are shown inside each node (ACTIVE: 1.0; DESENSITIZED: 0.5; INTERNALIZED: 0.1; UPREGULATED: 1.2). The state machine is deterministic with threshold-triggered transitions; see §3.4 for the relationship to a continuous-time Markov chain formulation.

3.4.2 Accumulator variables driving transitions

Three accumulator variables, evolving continuously between transition events, govern when transitions are eligible to fire:

- **Instantaneous saturation** as defined by eq. (20), expressing current ligand-binding occupancy.

- **Cumulative exposure trace** as defined by eq. (23), a leaky integral of saturation with default time constant τ .
- **Time-in-state**, an integer-incremented counter (in timestep units) measuring the duration since the most recent state transition. resets to zero whenever changes.

Together, these three variables capture the regulatory machinery's effective inputs: the current binding level (does the receptor's environment justify state change *now?*), the recent history of binding (has the cumulative drug-receptor interaction been large enough to push trafficking past a threshold?), and the residence time in the current state (has the receptor had enough time in this state to reach the next regulatory checkpoint?). Each transition in the framework depends on a specific subset of these accumulators, reflecting the corresponding biological mechanism.

3.4.3 Transition rules

The framework specifies six directed transitions between the four states. Each transition has an eligibility condition combining one saturation threshold, one cumulative or time threshold, and the requirement that the receptor be in the source state. When the eligibility condition is met, the transition fires deterministically; the rate kernel of the underlying CTMC notation is a step function with value 1 above threshold and 0 below. Default threshold values, listed below, were calibrated to match the canonical timescale separations documented in receptor-regulation reviews (Pierce et al., 2002; Gainetdinov et al., 2004) and are configurable per-subtype.

ACTIVE → DESENSITIZED (overstimulation). Fires when occupancy is sustained above a high threshold:

with defaults θ_H and τ_H . The sustained-time requirement prevents desensitization from triggering on brief phasic bursts and corresponds biologically to the time required for GRK phosphorylation and arrestin binding to accumulate.

DESENSITIZED → INTERNALIZED (prolonged exposure). Fires when cumulative exposure exceeds a threshold:

with default τ_I . The internalization decision is integrative rather than instantaneous because clathrin-mediated endocytosis is a downstream event of accumulated arrestin recruitment over time.

DESENSITIZED → ACTIVE (recovery). Fires when occupancy drops below a lower threshold and the receptor has spent a minimum recovery period in the desensitized state:

with defaults τ_{int} and τ_{up} . This corresponds to arrestin dissociation and re-coupling of the G-protein once ligand stimulation has subsided.

INTERNALIZED → ACTIVE (recycling). Fires after a minimum residence time, independent of current saturation:

with default τ_{int} . The pure-time dependence reflects the predominantly passive nature of receptor recycling, which proceeds on a timescale set by endosomal sorting machinery rather than by current ligand availability.

ACTIVE → UPREGULATED (understimulation). Fires when occupancy is sustained below a low threshold:

with defaults τ_{int} and τ_{up} . The longer sustained-time threshold relative to τ_{int} reflects the slower transcriptional and trafficking machinery that produces receptor supersensitivity.

UPREGULATED → ACTIVE (return to baseline). Fires when occupancy rises above a threshold and the receptor has spent a minimum residence in the upregulated state:

with defaults τ_{int} and τ_{up} .

3.4.4 State-entry effects on signaling parameters

A state transition is not just a change in τ_{int} — it is accompanied by adjustments to the other signaling parameters τ_{up} and τ_{act} that reflect the underlying biological mechanisms. On entry to each state, the framework applies multiplicative updates:

Transition target	Density factor	Sensitivity factor
DESENSITIZED	—	
INTERNALIZED		
UPREGULATED		
ACTIVE (recovery)	—	(passive recovery; see §3.4.5)

The density adjustment on internalization captures the fact that internalized receptors no longer count toward surface density. The compounding of τ_{int} , τ_{up} , and τ_{act} adjustments produces the empirically reasonable property that the effective signaling efficacy of an internalized receptor is multiplicatively small ($\sim 10^{-3}$ times the ACTIVE baseline, before accounting for τ_{int} and τ_{up} — internalization is reached only via the desensitized state, so the internalization factor compounds with the prior desensitization -halving), consistent with the near-quiescent signaling of fully internalized populations.

3.4.5 Sensitivity recovery and G-protein coupling dynamics

Two state-independent regulatory processes operate continuously alongside the discrete state transitions of §3.4.3.

Sensitivity recovery. When the receptor is in the ACTIVE state, sensitivity is restored linearly toward unity at a slow rate:

with default rate per unit time. This models the passive recovery of receptor sensitivity after desensitization, dephosphorylation and the restoration of unmodified intrinsic responsiveness.

G-protein coupling efficacy dynamics. The coupling efficacy is governed by a two-rate dynamics depending on whether saturation is above or below a degradation threshold :

with defaults , , , and floor . The asymmetric rates () reflect the biological observation that coupling efficacy degrades faster under sustained stimulation than it recovers under quiescence (Pierce et al., 2002). The floor prevents complete uncoupling, modeling the residual basal coupling that all signaling-competent receptors retain even under maximal desensitization.

3.4.6 Emotion-modulated plasticity

The framework supports a per-receptor plasticity layer in which sustained emotional or affective signals can produce additional adjustments to and beyond those imposed by the state machine of §3.4.4. Concretely, when an emotion signal exceeds a per-receptor threshold for a sustained period, the framework applies multiplicative or additive updates to and encoded in a per-receptor plasticity table. This mechanism is what allows the substrate to encode persistent changes (for example, the receptor-level traces of chronic stress, persistent affective state, or repeated reward associations) that outlast the immediate stimulation that produced them. The full plasticity-recipe table is provided in Supplementary §S3; we do not enumerate it here because (a) the table is large, (b) the specific values reflect engineering choices about how upstream emotional states should map to receptor parameter changes rather than analytical predictions, and (c) the framework's main quantitative claims hold for any choice of plasticity table satisfying basic monotonicity and boundedness conditions.

3.4.7 Homeostatic subtype switching

A final regulatory mechanism operates not within a single receptor but across the population of receptors for a given transmitter system: under sustained imbalance, the framework permits *subtype switching*, in which the relative densities of receptor subtypes for the same transmitter are reallocated to compensate for chronic over- or under-stimulation of specific subtypes. This is the framework's analog of the homeostatic plasticity documented at the synaptic level (Turrigiano & Nelson, 2004),

generalized to receptor-subtype proportions. The mechanism is specified by a set of directed switch rules; for an ordered subtype pair of the same transmitter, density transfers from to when 's exposure trace exceeds a threshold:

Correction (cross-paper inconsistency): the formula above is a stale draft, superseded by the corrected form given in the companion CTMC paper (§6.1), which was written to explicitly correct this exact reading. As printed above, r_{ab} multiplies $\rho^{(a)}(t)$ directly, making the transfer proportional to the current source density — this gives exponential decay, $\rho^{(a)}(t) = \rho^{(a)}(0)e^{-r^{(ab)}t}$, converging only asymptotically. The companion paper is explicit that the actual mechanism is constant-rate, not proportional: the transfer per step is $r_{ab} \cdot dt$ capped at $\delta_{\max}$ and by the remaining source density, independent of $\rho^{(a)}$ until that cap becomes binding:

Under this corrected form, density decays linearly (constant rate) rather than exponentially and reaches the absorbing state in finite time, exactly as derived and proven stable in the companion paper's §6.2–6.3. The conservation property stated in the paragraph below (every transfer subtracts from one subtype exactly what it adds to another) holds identically under both forms and is unaffected by this correction. The formula above should be read as superseded by this corrected form.

with per-pair transfer rate , per-step cap , and switch threshold (default in exposure units). Because every transfer subtracts from one subtype exactly what it adds to another, total receptor density for the transmitter system is conserved by construction; only the subtype proportions are redistributed. Subtype switching is the slowest of the framework's regulatory mechanisms, operating on timescales an order of magnitude longer than the per-receptor state transitions of §3.4.3, and a full analysis of its fixed points and stability is developed in a companion paper. The transfer rule above is, however, complete as stated, and the predictions in this paper that invoke subtype switching rest only on it and its conservation property, not on that further analysis.

Together, the mechanisms of this section specify the regulatory dynamics that turn the binding occupancy of Section 3.3 into time-varying state. The next section closes the loop by describing how the system's oscillatory state itself modulates the kinetic parameters that drive these dynamics.

3.5 Oscillatory layer

The two preceding sections describe a substrate of concentrations and receptor states that evolves on chemical and trafficking timescales. The third dynamical layer, developed in this section, operates on a faster timescale and a different domain: brain rhythms. We make oscillatory dynamics an explicit part of the framework rather than treating them as an emergent property of an underlying spiking network for three reasons. First, brain rhythms are not epiphenomenal — they modulate receptor affinity,

neurotransmitter release, and signal-to-noise ratios on timescales different from chemical clearance, and any framework that omits them loses a layer of biologically real dynamics (Buzsáki, 2006; Buzsáki & Wang, 2012). Second, oscillations provide the temporal scaffold against which cross-frequency coupling and theta-gamma binding occur, both of which are increasingly understood as central to learning and memory (Lisman & Jensen, 2013; Canolty & Knight, 2010). Third, an explicit oscillatory layer lets the framework express the bidirectional coupling between neurochemistry and rhythms — neuromodulator concentrations shape band amplitudes, and band amplitudes back-modulate the kinetic parameters that govern neurochemistry. The closure of this loop, developed at the end of this section, is one of the framework’s distinguishing features relative to prior computational neuroscience models of neuromodulators.

3.5.1 *The six oscillation bands and their functional roles*

The framework tracks six amplitude envelopes corresponding to the canonical EEG bands plus the sleep-spindle band:

Band	Frequency (Hz)	Default functional role	Primary citation
Delta ()	0.5 – 4	Recovery and reset, slow inhibition	Steriade et al. (1993)
Theta ()	4 – 8	Episodic binding, temporal sequence	Buzsáki (2002)
Alpha ()	8 – 12	Inhibitory gating, noise suppression	Klimesch (2012)
Beta ()	13 – 30	Precision, attentional control, contradiction detection	Engel & Fries (2010)
Gamma ()	30 – 100+	Binding, integration	Buzsáki & Wang (2012)
Sigma ()	12 – 15	Sleep spindles, thalamocortical replay (sleep-only)	Steriade (2006)

The framework represents each band as an amplitude envelope rather than as an oscillating waveform with explicit phase, on the grounds that the kinetic modulation rules of §3.5.4 depend principally on instantaneous amplitude rather than phase. The framework also tracks phase angles for each band (needed for any future extension to phase-amplitude coupling experiments) but the equations below depend only on the amplitudes. Sigma is sleep-restricted in our default configuration; during wakefulness, the closure rule of §3.5.2 produces near-zero sigma amplitude through the combined

effect of low GABA-tonic and high NE-tonic, without requiring an explicit wake/sleep state variable.

3.5.2 Neurotransmitter-to-band closure

The framework’s central oscillatory contribution is an explicit closure that derives each band’s amplitude from the current neurotransmitter state. For five of the six bands the amplitude is a clamped weighted sum of selected concentration components (the alpha band is the single exception, taking the conjunctive form given below):

where C_{band} may be the tonic component C_{tonic} , the phasic component C_{phasic} , or the total C_{total} , depending on the rule for band $band$. The default closure rules are summarized in the following table, with each entry $w_{band,transmitter}$ giving the weight on the indicated concentration component for the given transmitter–band pair:

Band	Transmitter contributions	Biological motivation
Gamma		Fast PV-driven gamma is gated by glutamatergic excitation and cholinergic modulation; phasic dopamine bursts boost gamma during reward and learning (Bartos, Vida, & Jonas, 2007; Lee et al., 2013).
Theta		Theta is driven primarily by septal cholinergic input (Buzsáki, 2002; Vertes & Kocsis, 1997), with cholinergic tone positively tracking theta power in vivo (Zhang, Karadas, et al., 2024), and is dopaminergically gated through the hippocampal–VTA loop (Lisman & Grace, 2005).
Alpha	(conjunctive; see below)	Alpha is the GABA-mediated inhibitory-gating rhythm (Klimesch, 2012), but specifically a rhythm of <i>relaxed wakefulness</i> —

Band	Transmitter contributions	Biological motivation
		abolished in sleep and desynchronized by active engagement. A term linear in tonic GABA alone mis-keys it to sleep (where tonic GABA is highest); the conjunctive form requires both inhibitory tone and waking noradrenergic arousal (Klimesch, 2012; Bazanova & Vernon, 2014), with a phasic-glutamate desynchronization term. Earlier 5-HT and CB1 terms were removed on review (Appendix C).
Beta		Beta tracks adaptive gain and attentional control, driven by locus-coeruleus noradrenaline (Aston-Jones & Cohen, 2005).
Delta		Delta-dominated slow-wave states are pro-inhibitory; endogenous opioid and cannabinoid tone are pro-delta sedatives.
Sigma		Thalamocortical spindles depend on GABAergic TRN bursting and glutamatergic relay rebound, and are actively suppressed by locus-coeruleus NE during waking (Steriade et al., 1993; Steriade, 2006).

Five of the six bands take the linear form of eq. (32); alpha is the single exception. Because alpha is a rhythm of *relaxed wakefulness* (present when cortex is inhibited *and* awake, but abolished in sleep and suppressed during active sensory processing), a term linear in tonic GABA alone mis-specifies it, since tonic GABA is in fact highest

during slow-wave sleep. We therefore give alpha a conjunctive closure that gates inhibitory tone by waking noradrenergic arousal and subtracts a phasic-glutamate desynchronization term:

with defaults $\alpha_{\text{w}} = 0$ and $\alpha_{\text{d}} = 0$. The product term encodes the conjunction — alpha appears only when both inhibitory tone (GABA) and waking arousal (NE) are present, and collapses in NREM as locus-coeruleus output falls — while the phasic-glutamate term captures alpha desynchronization during active engagement. This is the one place the closure departs from strict linearity; we return to its motivation, and to the limitation it still carries (the arousal dependence is monotonic, so only the sleep–wake limb of alpha’s inverted-U dependence on arousal is captured), in §6.2.

The weights are the framework’s defaults and are configurable per-band. The negative weight on NE-tonic for the sigma band is biologically motivated: noradrenergic suppression of thalamocortical spindles is a well-characterized mechanism for keeping spindles confined to NREM sleep (Steriade, 2006). It also explains, mechanistically, why the framework does not require an explicit sleep flag — high NE-tonic during waking drives σ to zero through the closure rule itself.

The closure represented by eq. (32) and the table above is the central object of Section 3.5: a fully specified mapping from a 12-dimensional concentration vector to a 6-dimensional band amplitude vector, with each entry biologically motivated. It is important to position this correctly against existing work. Generative models of EEG band activity are well established — the neural-mass and neural-field lineage, and in particular the dynamic causal modeling (DCM) family (Jansen & Rit, 1995; David & Friston, 2003; Moran et al., 2007), produces band-specific spectra from biophysical population dynamics, and neuromodulatory influence on those spectra has been modeled explicitly, including cholinergic modulation (Moran et al., 2013). Those models generate oscillations *dynamically*, as emergent properties of simulated interacting excitatory and inhibitory populations, with neuromodulators and receptors entering as parameters that reshape the simulated dynamics. What the present framework does differently is not to claim that the chemistry-to-rhythm link is new, but to express it as a *direct algebraic closure*: a closed-form map from neurotransmitter concentrations straight to band amplitude (linear in the concentrations for five of the six bands, with a single conjunctive term for the alpha band) with no intervening population-dynamics simulation. This is a more reductive construction than a neural-mass model (it trades mechanistic depth for closed-form tractability and for direct couplability to the concentration and receptor layers of the framework) and its novelty is precisely that tractable directness, not the existence of a chemistry-to-band relationship. We are not aware of prior work that specifies the neuromodulator-to-band mapping in this direct, closed-form way; we make that limited claim rather than the stronger and false claim that chemistry-driven band models do not exist. Generative neuromodulator-driven models of band activity do exist, including thalamocortical models that drive sleep-stage spectra by varying neuromodulator-linked parameters, and a verbal

neuromodulator-balance hypothesis for EEG rhythms predates this work; we discuss both, and the precise residual our construction retains against them, in §4.5.4.

3.5.3 Cross-frequency coupling envelopes

The framework computes three cross-frequency coupling envelopes as products of band amplitudes:

Each product captures a functional regime documented in the cross-frequency coupling literature: theta-gamma coupling indexes associative encoding and the temporal binding of distributed representations (Lisman & Jensen, 2013); alpha-beta coupling indexes impulse suppression and inhibitory control (Engel & Fries, 2010); delta-sigma coupling indexes NREM consolidation, where the slow-oscillation and spindle phases coordinate memory replay (Steriade, 2006; Mölle & Born, 2011). The framework treats CFC envelopes as derived quantities rather than independent state variables, which has the desirable property that they cannot be non-zero unless both constituent bands are non-zero — a constraint that follows automatically from the multiplicative form. The default product form, $C_{\alpha\beta}$, is the simplest non-trivial coupling; the framework supports more general functional forms (e.g., gated products, phase-amplitude couplings) for extensions that require them.

3.5.4 Oscillatory modulation of kinetic parameters

Band amplitudes back-modulate six classes of kinetic parameters introduced in earlier sections, closing the bidirectional loop. The default modulation rules are single-band and linear; the framework supports multi-band variants with band-specific coefficients summed across all bands, but the single-band forms below are used as defaults.

Receptor affinity (K_d). Theta-band activity lowers the dissociation constant, increasing apparent binding affinity (Section 3.3.2):

with default K_d and a floor at $K_{d,\min}$ to keep K_d positive. Biologically, theta-band modulation of dopaminergic affinity has been particularly well-characterized; the framework generalizes this single-band default and supports per-receptor band-coefficient tables (Appendix E).

Release rate. Gamma boosts phasic release (eq. 13 with the gamma coupling term as the dominant contribution):

with default $\gamma = 1$. This captures the documented gain that gamma-band activity provides to phasic vesicular release. Eq. (37) is the shipped single-band default of eq. (13)'s general multi-band form — the special case where the gamma coefficient carries the coupling and every other $\beta_{k,i}$ is zero — not a competing or superseding definition; readers wiring in additional bands via Appendix E's per-receptor tables are extending eq. (37) back toward the general eq. (13) form, not choosing between two incompatible rules.

Noise amplitude. Alpha suppresses stochastic noise, modeling the documented role of alpha in inhibitory gating and dynamic stabilization:

with default $\alpha = 1$. The floor at α_{floor} prevents the noise coefficient from reaching zero — a zero-noise state would collapse the SDE to a deterministic ODE and could lock the system on a fixed point, eliminating the endogenous variability the framework relies on for stochastic exploration of state space. The floor preserves a minimum of stochastic exploration even under maximal alpha suppression.

Reuptake rate. Beta boosts transporter activity, tightening clearance dynamics:

with default $\beta = 1$. This couples attentional/precision states (high beta) to faster neurotransmitter clearance, consistent with the empirical association between beta-band activity and tightened sensorimotor control.

Tonic baseline. Delta lowers the tonic baseline, modeling the documented hypoactivation of monoaminergic systems during slow-wave activity:

with default $\delta = 1$ and a floor at δ_{floor} . The modulated baseline is the reversion target for the tonic component (eq. 4); high delta produces a temporarily lowered baseline that the system reverts toward, reproducing the lowered baseline tone characteristic of slow-wave states.

Receptor state-transition timing. Oscillatory state modulates the effective time-thresholds of the receptor regulatory state machine (Section 3.4) through a multiplier τ :

with non-negative band coefficients τ_k . The default coupling is beta-band acceleration of desensitization with coefficient τ_β , capturing the empirical observation that high-precision/high-vigilance states accelerate the kinetics of receptor regulation. Eq. (41) is the most general of the six modulation rules and accepts arbitrary band-coefficient tables for per-receptor specialization.

3.5.5 The closed-loop dynamics

Equations (32) through (41) close the framework's dynamical loop. Reading the loop forward: neurotransmitter concentrations drive band amplitudes through eq. (32); band amplitudes back-modulate the kinetic parameters μ , ν , λ , and the receptor state-transition timing through eqs. (36–41); the modulated parameters change the evolution of $\mathbf{y}$ on the next timestep, which changes $\mathbf{y}$, which changes the parameters again, and so on. The system is therefore not a feed-forward map from upstream cognitive signals to receptor signaling but a closed coupled-dynamical system in which neurochemistry and rhythms co-evolve.

Several qualitative properties follow from this structure. First, the system supports positive-feedback loops on multiple bands: a phasic dopamine burst increases gamma, which increases dopamine release through eq. (37), which sustains gamma. This loop is bounded by fatigue (eq. 15), receptor desensitization (Section 3.4), and the multiplicative noise floor of eq. (7), which prevent runaway dynamics. Second, the system supports cross-band antagonism: high alpha suppresses noise and limits the stochasticity needed for transitions out of stable states, while high beta accelerates state transitions through eq. (41), producing a dynamic in which different attentional regimes (alpha-dominated vs. beta-dominated) yield quantitatively distinct receptor regulatory behavior. Third, the system supports state-dependent timescale separation: when bands are at characteristic levels for a given regime (e.g., NREM sleep with high delta, low NE, and emergent high sigma), the modulation rules collectively shift the system into a qualitatively different operating regime, reproducing the homeostatic and consolidation-relevant dynamics that motivate the inclusion of the oscillatory layer in the first place.

We return to specific instances of these emergent behaviors in Section 4 (Model behavior), where each is grounded in a concrete worked example.

3.6 The coupled system

Sections 3.2 through 3.5 specify four dynamical layers of the framework: stochastic kinetics for neurotransmitter concentrations, deterministic binding and effective signaling at receptors, a regulatory state machine on receptor functional state, and an oscillatory closure that derives band amplitudes from the underlying concentration vector and back-modulates the kinetic parameters of the lower layers. These four layers do not operate independently. They form a single coupled dynamical system in which a perturbation introduced at any layer propagates to the others on its own timescale, and in which qualitative regime properties (the resting-state band profile, the cascade trajectory under sustained drive, the emergence of cross-frequency coupling envelopes) emerge from the interaction of the layers rather than from any layer alone.

The framework's coupling loop is closed and bidirectional. Reading it forward: cognitive signals drive release rates through eq. (8), modulated by fatigue (eq. 12) and oscillatory state (eq. 13); release rates enter the SDE drift of eqs. (4–5), which together with the square-root multiplicative noise of eq. (7) evolve the concentration components $\mathbf{x}$ and $\mathbf{y}$;

the concentration components determine the localization-weighted effective concentration via eq. (19), which sets receptor occupancy via eq. (20); occupancy and the receptor state variables together determine the effective signaling proxy via eq. (22) and the exposure trace via eq. (23); the exposure trace and saturation jointly drive the regulatory state machine of §3.4, which updates ρ , σ , γ , and $g(\chi)$; and finally the concentration vector drives the oscillatory closure of eq. (32), which back-modulates dissociation constants (eq. 36), release rates (eq. 37), noise amplitudes (eq. 38), reuptake (eq. 39), tonic baselines (eq. 40), and state transition timing (eq. 41). The loop closes because the modulated parameters re-enter the SDE evolution of the next timestep, completing the cycle.

The architecture's distinctive character is its *multi-timescale structure*. Concentration dynamics evolve on the timescale of the integration step and the OU reversion rate; oscillatory amplitudes co-evolve with concentrations through the closure of §3.5.2; receptor regulatory transitions occur on slower timescales set by the threshold conditions of §3.4.3 (seconds for desensitization, tens to hundreds of timesteps for internalization and recycling); and subtype switching, when active, operates on the slowest timescale of all. The coupled system therefore produces phenomena whose characteristic timescales are not set by any single parameter but emerge from the interplay between layers.

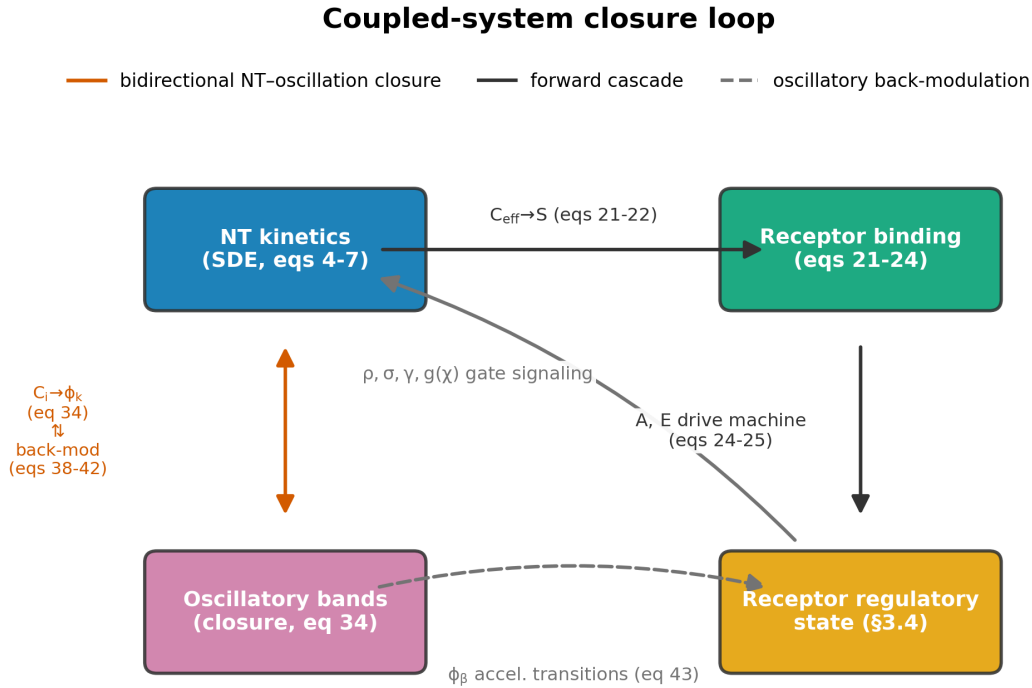


Figure 3.6: Coupled-system closure loop

Figure 3.6. Coupled-system closure loop. Four dynamical layers (NT kinetics, receptor binding, receptor regulatory state, oscillatory bands) and their directional couplings. Forward cascade (black arrows): NT concentrations drive receptor saturation (eqs. 19–20); activation and exposure drive the regulatory state machine (eqs. 22–23). Bidirectional NT–oscillation closure (orange double-headed arrow): via the closure rules (eq. 32); back-modulation of NT kinetic parameters via eqs. 36–40. Regulatory feedback (gray diagonal): gate downstream signaling. Oscillatory back-modulation of transitions (gray dashed arrow): accelerates receptor state transitions (eq. 41). The closure loop is the defining structural feature of the framework: each layer’s outputs are inputs to at least one other layer, producing emergent multi-timescale dynamics unavailable from any single layer in isolation.

The qualitative properties of the resulting system (boundedness under all default parameter regimes, multi-stability across functional regime configurations, emergent oscillatory signatures, and multi-timescale habituation under sustained drive) are the subject of the next section, which works through specific worked examples that demonstrate each property.

4. Model behavior

4.1 Steady-state behavior under tonic input

The mean-reverting structure of the drift term, eq. (4), predicts that under zero release drive and constant loss parameters the tonic component of any transmitter system converges to a stationary distribution centered at its baseline concentration. For an isolated Ornstein–Uhlenbeck process with constant diffusion coefficient, the analytical mean and variance are:

so that the trajectory approaches the baseline at rate λ and fluctuates around it with stationary standard deviation σ . The full system, with the multiplicative noise of eq. (7) and the multi-channel loss of eq. (6), does not admit a closed-form stationary distribution in general; nevertheless eq. (42) provides a useful first-order prediction for the mean, and eq. (43) gives an upper bound on the variance over the operating range.

We verify these predictions numerically with a canonical test configuration: $\mu = 1$, $\lambda = 1$, constant diffusion $D = 1$, step size $\Delta t = 0.01$, integration horizon $T = 100$, with reuptake/degradation/diffusion channels disabled to isolate the OU behavior of the drift term. Under these parameters, eq. (42) predicts $\mu_{ss} = 1$, and eq. (43) predicts a stationary standard deviation of $\sigma_{ss} = 1$. Numerical integration over 100 independent seeded runs produces an empirical mean at $\mu_{ss} = 1$ and an empirical standard deviation $\sigma_{ss} = 1$, matching the analytical predictions to within Monte Carlo error.

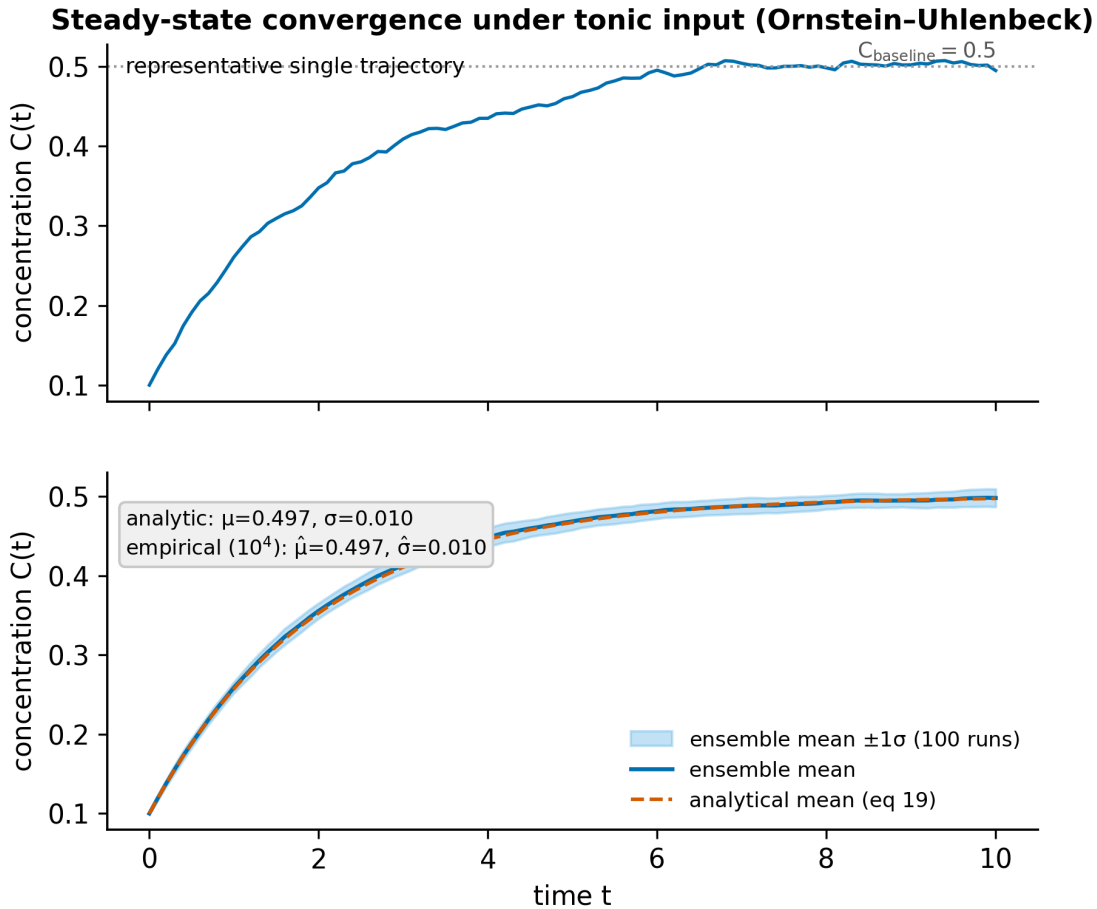


Figure 4.1: Steady-state convergence under tonic input

Figure 4.1. Steady-state convergence under tonic input (Ornstein–Uhlenbeck).

Parameters: , , , , . *Top panel:* representative single trajectory. *Bottom panel:* ensemble mean (solid blue) and band (100 runs) overlaid on the analytical mean curve (dashed, eq. 42). Inset: analytical stationary mean and sd confirmed against an independent-seed empirical ensemble (seed 20240615: ,), closing the \$4.1 claim. The band is narrow because the stationary standard deviation is genuinely small; this is not a rendering artifact.

The result confirms that the integrator satisfies the asymptotic Ornstein–Uhlenbeck behavior predicted by the drift structure, validates the seeded-RNG determinism of the implementation, and provides a quantitative anchor for the more complex multi-channel and multi-NT dynamics analyzed in subsequent sections.

4.2 Phasic burst dynamics

A second canonical model behavior is the response to discrete release events: a transient phasic burst that rises rapidly, saturates as a function of drive strength, and decays exponentially back to the tonic baseline. We characterize this behavior in two

parts: the analytical structure of an isolated burst (§4.2.1) and the modulation of repeated bursts under accumulated fatigue and adaptive thresholding (§4.2.2).

4.2.1 Isolated burst trajectory

A release event at time t produces an instantaneous phasic increment whose amplitude is set by the saturating exponential of eq. (14), restated here for reference:

After the impulse, the phasic component decays according to its OU dynamics, eq. (5), with reversion target zero and reversion rate λ . The deterministic envelope of the decay is therefore:

where λ_{eff} is the linearized loss rate at low concentrations (the leading-order contribution from the three clearance channels of eq. 6, evaluated near zero). The effective decay rate is therefore the sum of the OU reversion rate and the linearized loss rate, and the half-life of the burst is τ_{eff} .

Two analytical properties of eq. (44) are worth highlighting because they constrain the framework's qualitative behavior. First, the burst amplitude is **bounded above** by A_{max} regardless of how large u becomes; the implementation verifies this property explicitly with high-drive inputs (e.g., $u = 10$ yields $A \approx 0.99$ within machine precision). This is the framework's first line of defense against runaway dynamics under pathological upstream signaling. Second, the burst response is **near-linear in drive** at low drive levels, where $u \ll 1$ for $\lambda_{\text{eff}} \ll \lambda$, transitioning to saturation only as u approaches unity. A low drive of $u = 0.1$ with default sensitivity $\lambda = 1$ yields $A \approx 0.1$, while a moderate drive of $u = 1$ yields $A \approx 0.5$, and a high drive of $u = 10$ yields $A \approx 0.99$. The framework therefore preserves linear coding in the typical operating range while saturating gracefully under extreme inputs, mirroring the empirically documented dose-response curves of vesicular release.

Combining the deterministic envelope of eq. (45) with the multiplicative noise of eq. (7) gives the full stochastic burst trajectory: a saturated amplitude impulse followed by an OU mean-reverting decay with state-dependent noise. The noise envelope is largest near the peak of the burst (where A is large) and shrinks as the trajectory returns to baseline, reproducing the empirical observation that variability in phasic responses scales with the response magnitude.

Isolated phasic burst — decay rate λ illustrative; structure (saturation, exp. decay, $\sqrt{C}$ noise) is the claim

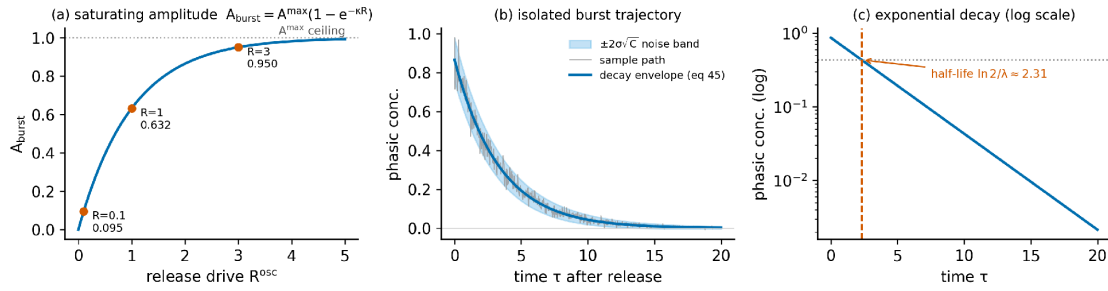


Figure 4.2.1: Isolated phasic burst

Figure 4.2.1. Isolated phasic burst. The decay rate is illustrative; the documented structural claims are saturation, exponential decay, and noise scaling. (a) Saturating amplitude function (eq. 14) over release drive, with reference points at . (b) Single burst trajectory: instantaneous rise to at , exponential decay envelope (eq. 45, solid), and noise band showing the CIR-style multiplicative noise scaling that produces heteroscedastic spread proportional to amplitude. (c) Log-scale view of the decay phase confirming the straight-line exponential structure; dashed vertical line marks the predicted half-life .

4.2.2 Repeated bursts: fatigue and adaptive thresholding

When release events occur in succession, three mechanisms collectively reshape the burst train.

Fatigue accumulation. Each burst contributes to the fatigue variable through eq. (15), which accumulates with activity and decays passively. Above the threshold, fatigue actively suppresses subsequent release through the gating function of eq. (12), so a sufficiently rapid burst train produces a sequence of decreasing-amplitude bursts. Fatigue additionally slows the reversion rate through eq. (16), meaning that the bursts produced under high fatigue not only have smaller amplitudes but also decay more slowly than fresh-system bursts.

Adaptive thresholding. The release-drive computation includes a recent-activity trace that raises the effective threshold for novelty and effort drives:

where is an exponentially decaying moving average of recent release drive with time constant in default units. This is a second, faster habituation mechanism distinct from fatigue: where fatigue suppresses release machinery, the adaptive threshold suppresses release initiation, producing a regime in which the first burst in a sequence may pass the threshold cleanly while subsequent stimuli of the same intensity are increasingly likely to be filtered.

Receptor desensitization. A third mechanism, developed in Section 3.4, operates downstream of release: sustained occupancy above θ_{desens} drives the receptor population into the DESENSITIZED state, where θ drops to θ_{desens} and the effective signaling per unit binding is halved. Repeated bursts on the same receptor population therefore produce a progressive attenuation in downstream signaling even when the bursts themselves remain at constant amplitude. The cumulative exposure trace of eq. (23) then drives further transition to INTERNALIZED if the train continues long enough.

The three mechanisms act on distinct timescales: adaptive thresholding on the order of the burst-train spacing (tens of timesteps), fatigue on a slower scale (tens to hundreds of timesteps), and receptor regulatory state changes on yet slower scales (the time constants of §3.4.3). Together, they make the framework's response to repeated stimulation quantitatively distinct from its response to isolated stimulation in three observable ways: amplitude suppression, decay-time prolongation, and downstream signaling attenuation. We return to the downstream attenuation in Section 4.3, which traces the full receptor adaptation cascade under sustained drive.

Constants illustrative; the three documented mechanisms (fatigue, adaptive threshold, desensitization) are the claim

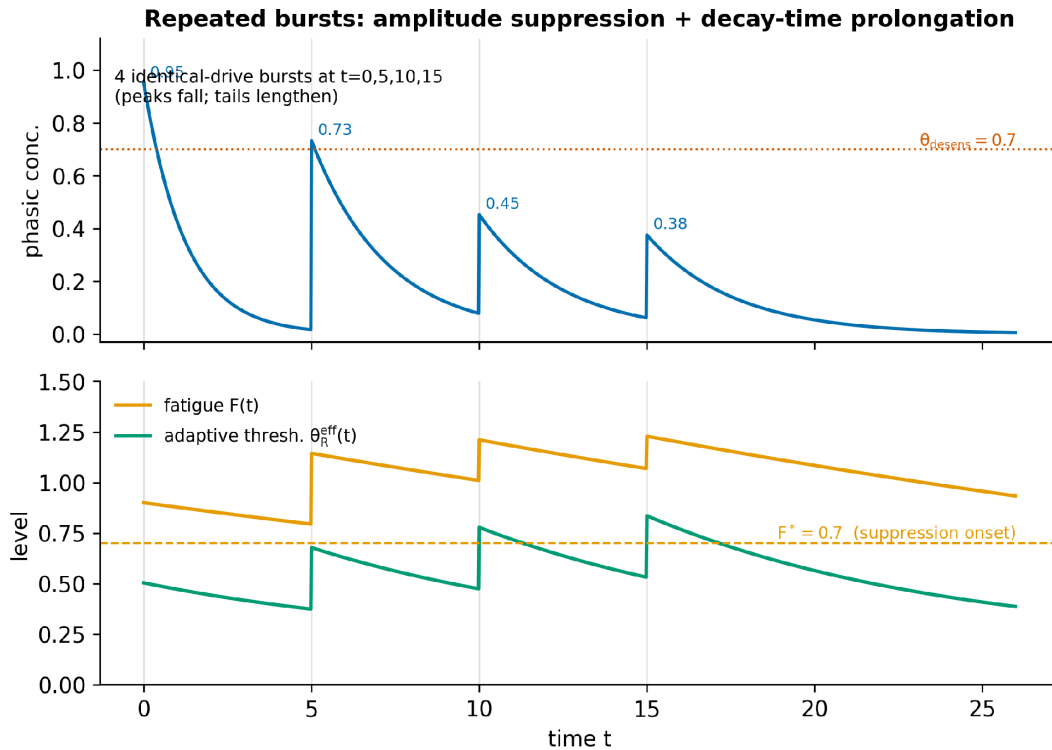


Figure 4.2.2: Repeated bursts — amplitude suppression and decay-time prolongation

Figure 4.2.2. Repeated bursts: amplitude suppression and decay-time prolongation. Four identical-drive bursts at $t=0, 5, 10, 15$. Constants are illustrative; the three documented mechanisms are the claim. *Top panel:* phasic concentration trajectory. Peak amplitudes fall monotonically ($0.95 \rightarrow 0.73 \rightarrow 0.45 \rightarrow 0.38$) as fatigue crosses F^* from the second

burst onward; inter-burst decay times lengthen (burst 1 returns to baseline by ; burst 4 persists past) as fatigue slows the reversion rate (eq. 16). Dotted line: . *Bottom panel:* fatigue variable (orange) ratcheting upward across the train; adaptive threshold (green) rising in parallel, with each burst adding to the running average that raises the effective release threshold. Both effects operate simultaneously with receptor desensitization (§4.3), producing the three-mechanism habituation profile described in §4.2.

The phasic burst behavior described here is the framework’s transient response to discrete release events. The next section traces what happens to the receptor population when bursts of this type are sustained long enough to drive the full regulatory cascade.

4.3 Receptor adaptation cascade

The phasic burst dynamics of Section 4.2 describe the transient response to discrete release events; here we analyze the framework’s behavior when stimulation is sustained long enough to drive the receptor population through the full regulatory cascade described in Section 3.4. The cascade has four distinct phases (a desensitization phase, a [conditional] internalization phase, a recovery phase, and a [conditional] upregulation phase) separated by the state transitions of eqs. (24–29). We walk through each phase analytically, then assemble the predicted trajectory.

4.3.1 Cascade trajectory under sustained high drive

We consider a receptor that begins in the ACTIVE state, with τ_{act} , τ_{des} , and a sustained input that maintains saturation above the desensitization threshold θ_{des} . The cascade unfolds in the following stages.

Stage 1: Onset of desensitization (ACTIVE → DESENSITIZED). The transition fires when the time-in-state exceeds τ_{act} with saturation above threshold (eq. 24). On entry to DESENSITIZED, sensitivity is multiplied by α , and the state-gate drops from τ_{act} to τ_{des} . The effective signaling proxy (eq. 22) therefore drops by a factor of α immediately on transition — half from the τ_{act} multiplier and half from the α change — independent of saturation level. With sustained τ_{act} , the framework predicts a step-down from τ_{act} in ACTIVE to τ_{des} on entry to DESENSITIZED. The $4\times$ drop in single-receptor effective signaling represents the framework’s prediction for the immediate post-desensitization phase.

Stage 2: Cumulative exposure and conditional internalization. While the receptor remains in DESENSITIZED with elevated τ_{des} , the exposure trace (eq. 23) accumulates according to its leaky-integrator dynamics. With time constant τ_{int} and bounded saturation τ_{int} , the trace approaches an asymptote τ_{int} . The default internalization threshold θ_{int} therefore lies above the asymptote of the default integrator under bounded saturation, with the consequence that internalization under default per-receptor parameters is a *conditional* event rather than a guaranteed one: it requires either per-receptor tuning that lengthens τ_{int} to allow higher asymptotes, or transient excursions in saturation associated with brief phasic events that push the integrator past the asymptote that bounded τ_{int} alone would produce. We treat this calibration choice as a deliberate framework property

(internalization is rare under sustained tonic stimulation and frequent under phasic-saturated regimes) and discuss the resulting predictions in §4.3.4 below.

Stage 3: Internalization (when reached). When ρ crosses ρ_{th} , the receptor transitions DESENSITIZED $\rightarrow$ INTERNALIZED (eq. 25). On entry, density is multiplied by γ , sensitivity by β , and the state-gate falls to g_{int} . The compound effect on the signaling proxy is dramatic: starting from the post-desensitization ρ , the receptor enters internalization with ρ_{int} for the same saturation ρ_{th} . The receptor is, for practical purposes, functionally silenced.

Stage 4: Recovery. Recovery from internalization is time-gated, not saturation-gated: the receptor returns to ACTIVE after spending τ_{int} timesteps in INTERNALIZED (eq. 27), regardless of the current ligand availability. This reflects the predominantly passive nature of clathrin-mediated trafficking and recycling. Recovery from desensitization (without intervening internalization) is instead saturation-gated: when ρ drops below ρ_{th} and the receptor has spent at least τ_{des} timesteps in DESENSITIZED, the transition DESENSITIZED $\rightarrow$ ACTIVE fires (eq. 26). On return to ACTIVE, sensitivity recovers passively toward β_{act} at rate γ_{act} per timestep (eq. 30), so a full recovery from a single β -halving requires roughly τ_{act} timesteps. The receptor is therefore not signaling-competent at full efficacy immediately on return to ACTIVE; full restoration takes additional time, during which the receptor is again sensitive to a fresh stimulation cycle.

The combined cascade trajectory is therefore characterized by three timescales: a fast desensitization transition at τ_{des} timesteps, a slow exposure-accumulation phase whose outcome depends on per-receptor parameter tuning, and a recovery phase whose duration depends on whether internalization was reached (passive 50-timestep recycling) or only desensitization (saturation-gated recovery plus ~ 14 timesteps of sensitivity restoration).

Receptor adaptation cascade (deterministic; thresholds from §3.4). Sustained tonic $S=0.8$ reaches DESENS but not INTERN (exposure asymptote $8 < 15$).

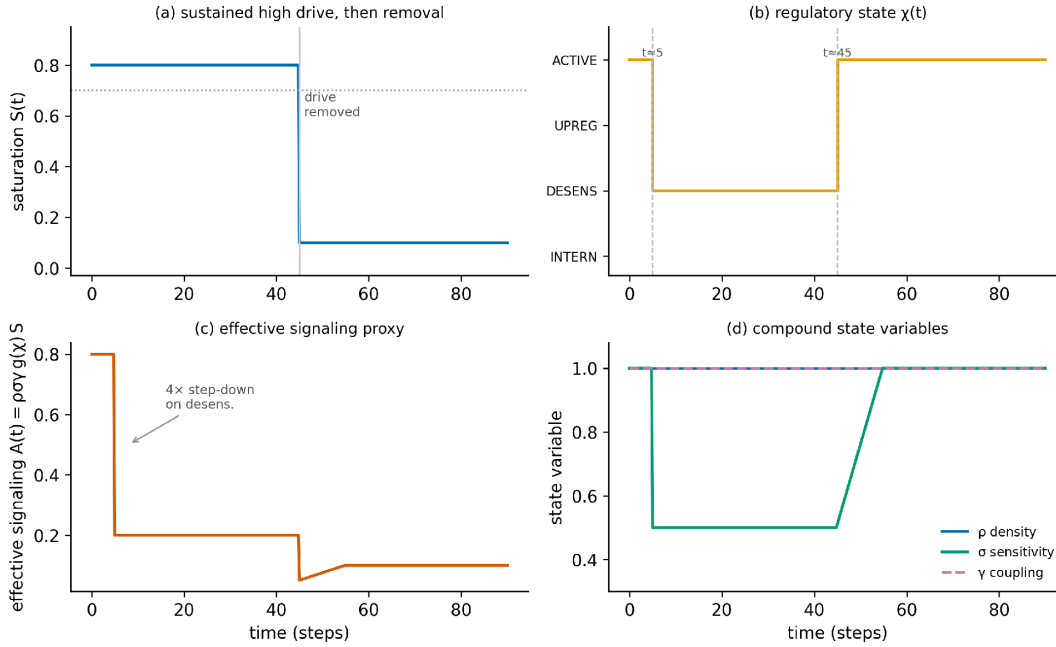


Figure 4.3.1: Receptor adaptation cascade

Figure 4.3.1. *Receptor adaptation cascade (deterministic; all thresholds from §3.4).*

Sustained tonic drive is applied from $t=0$ to $t=45$, then removed ($t=45$). (a) Saturation S ; dotted line at $S \approx 0.7$ (desensitization threshold). (b) Regulatory state: ACTIVE $\rightarrow$ DESENSITIZED at $t=5$ (when $S \geq 0.7$); DESENSITIZED $\rightarrow$ ACTIVE at $t=45$ (drive removal triggers recovery: $t=45$, steps). Sustained drive asymptote $S=0.8$, so INTERNALIZED state is never reached. (c) Effective signaling proxy: $4\times$ step-down at desensitization onset ($t=5$), further suppression at drive removal, then gradual recovery as S restores. (d) Compound state variables (density, flat), (sensitivity, dipping to 0.5 at desensitization, recovering under ACTIVE state), (coupling, flat).

4.3.2 Conditional upregulation under prolonged understimulation

The opposite branch of the cascade applies when drive is removed for an extended period. A receptor in ACTIVE with $S \geq 0.7$ for at least $t_{\text{upreg}}=15$ timesteps transitions to UPREGULATED (eq. 28). On entry, density and sensitivity are both boosted ($\rho=1.5$ and $\sigma=1.5$), and the state-gate jumps to $S=1.0$. The compound effect is an effective signaling boost of $2.25\times$ at the next time the receptor is stimulated — almost a $2\times$ supersensitivity. This is the framework's mechanistic correlate of the receptor-supersensitivity phenomenon observed clinically after sustained antagonist exposure or in withdrawal states (Pierce et al., 2002). Return to ACTIVE from UPREGULATED is fast and occurs when stimulation resumes (eq. 29).

4.3.3 Subtype switching under chronic imbalance

When a single receptor subtype is held in a chronic adaptation cycle for timescales an order of magnitude longer than the per-receptor regulatory transitions of §4.3.1, the

framework activates a population-level homeostatic mechanism: subtype switching reallocates total density across subtypes of the same transmitter system, reducing the density of the chronically desensitized subtype and increasing the density of less-stimulated subtypes, conserving total density (§3.4.7). The qualitative prediction this supports rests only on the transfer rule and conservation property already given in §3.4.7, not on any further analysis: under chronic-imbalance regimes, the framework predicts not just attenuated single-receptor signaling but a *redistribution* of receptor subtype expression — a pattern documented for several monoaminergic systems following sustained pharmacological perturbation. The framework’s fixed-point and stability analysis for this mechanism is developed in a companion paper, but that analysis is not needed to state or test the redistribution prediction itself.

4.3.4 Functional implications

The full cascade serves a clear regulatory function in the framework: it converts sustained high-drive inputs into progressively attenuated downstream signaling on three timescales, providing a multi-stage barrier against runaway signaling under pathological stimulation while preserving phasic response capacity under intermittent stimulation. The calibration gap noted in Stage 2 above (that default) keeps internalization a conditional event rather than a guaranteed one is itself a meaningful design property: the framework predicts that internalization is reached not by tonic saturation alone but by sustained phasic-saturated regimes, consistent with the empirical observation that chronic-drug receptor internalization correlates more with phasic peak exposure than with average tonic concentration. This is a non-trivial qualitative prediction that emerges from the interaction between the exposure-trace dynamics of eq. (23) and the saturation function of eq. (20), and would not arise from a simpler concentration-driven regulatory model. We return to this prediction in Section 5 (Discussion) as one of the framework’s hypothesis-generating outputs.

The receptor adaptation cascade described here completes the within-receptor regulatory story. The next section turns to the higher-order dynamics that emerge when many receptors of distinct types co-evolve under correlated transmitter inputs.

4.4 Oscillatory emergence from neurotransmitter co-fluctuation

The closure rules of Section 3.5 specify a deterministic mapping from a 12-dimensional concentration vector to a 6-dimensional band amplitude vector, but the rules alone do not demonstrate that the closure produces *recognizable* dynamics under realistic input regimes. In this section we walk through three input regimes (a resting-state baseline, a phasic burst event, and a sleep-like configuration) and show that the framework’s closure rules generate band-amplitude signatures qualitatively consistent with empirical EEG observations for the corresponding states. The three regimes are chosen for their representational distinctness; they are not the only regimes the framework can express, but they establish that the closure is doing meaningful work rather than producing degenerate or uninformative outputs.

4.4.1 Resting-state band signature

We consider a balanced tonic input regime corresponding to quiet wakefulness: $\tau_{\text{GABA}} = \tau_{\text{glutamate}} = \tau_{\text{cholinergic}} = \tau_{\text{dopaminergic}} = \tau_{\text{serotonergic}} = \tau_{\text{histaminergic}} = \tau_{\text{noradrenergic}} = \tau_{\text{thyrotrophic}}$, all other neurotransmitters at low baseline, and zero phasic components. Applying eq. (32) with the default closure rules of §3.5.2 yields the band-amplitude profile:

Band	Computation	Amplitude
Delta		
Theta		
Alpha		
Beta		
Gamma	(no phasic)	
Sigma		

The profile is alpha-dominant: alpha is highest (α), with theta just below (θ), beta lower (β), low delta and sigma, and zero gamma. This reproduces the canonical resting-state EEG signature of quiet eyes-closed wakefulness — the posterior alpha rhythm (the Berger rhythm) carrying the dominant resting power, with beta indexing vigilance and gamma minimal in the absence of task-driven engagement (Klimesch, 2012; Buzsáki, 2006). Reproducing alpha-dominant rest is a basic consistency check rather than a novel prediction, but it is one a purely linear, GABA-only closure cannot pass: keying alpha to tonic GABA alone leaves the resting profile theta-dominant, because tonic GABA is no higher in quiet wakefulness than the cholinergic-dopaminergic drive of theta. The conjunctive alpha closure (§3.5.2) resolves this by gating GABAergic inhibitory tone on waking noradrenergic arousal, so that alpha leads at rest and, as §4.4.3 shows, collapses correctly in NREM. The framework reproduces this signature *without an explicit “resting state” variable* — it emerges as a direct consequence of the tonic neurotransmitter levels and the closure rules.

4.4.2 Phasic-driven gamma emergence and closure-loop feedback

We next consider a transient phasic burst event superimposed on the resting baseline: $\tau_{\text{GABA}} = \tau_{\text{glutamate}} = \tau_{\text{cholinergic}} = \tau_{\text{dopaminergic}} = \tau_{\text{serotonergic}} = \tau_{\text{histaminergic}} = \tau_{\text{noradrenergic}} = \tau_{\text{thyrotrophic}}$, all tonic levels unchanged. Substituting into eq. (32) with the gamma closure rule yields:

a substantial gamma amplitude that emerges directly from the phasic NT components and would have been zero at rest. The framework therefore predicts a transient gamma signature concurrent with phasic dopaminergic-glutamatergic-cholinergic activity, consistent with the empirical association between gamma and salience/learning events (Bartos et al., 2007; Buzsáki & Wang, 2012).

The closure loop now becomes consequential. With $\tau_{\text{GABA}} = \tau_{\text{glutamate}} = \tau_{\text{cholinergic}} = \tau_{\text{dopaminergic}} = \tau_{\text{serotonergic}} = \tau_{\text{histaminergic}} = \tau_{\text{noradrenergic}} = \tau_{\text{thyrotrophic}}$ and the default release-modulation coefficient $\tau_{\text{GABA}} = \tau_{\text{glutamate}} = \tau_{\text{cholinergic}} = \tau_{\text{dopaminergic}} = \tau_{\text{serotonergic}} = \tau_{\text{histaminergic}} = \tau_{\text{noradrenergic}} = \tau_{\text{thyrotrophic}}$ from eq. (37), the next-timestep release rate is multiplied by:

a boost in release sensitivity that feeds back into the phasic components of DA, GLU, and ACh — partially sustaining the gamma elevation rather than letting it decay on the same timescale as the underlying NT phasic components. The framework therefore predicts that gamma events under closure-loop feedback are not impulse responses but bounded oscillatory pulses with characteristic shape: a fast rise driven by the underlying phasic input, a slower decay shaped by the closure-loop self-sustainment, and a hard ceiling imposed by the saturation function (eq. 14) and the desensitization cascade (Section 4.3). This shape is one of the framework’s specific qualitative predictions and would be testable against EEG-monitored learning paradigms.

4.4.3 Sleep-like configuration and the emergence of sigma

The third regime illustrates the framework’s ability to express qualitatively distinct band signatures under different tonic mixtures, without any sleep-specific state variable. We consider a configuration corresponding to NREM-analog dynamics: (elevated inhibitory tone), (collapsed locus-coeruleus output), (elevated endogenous opioid), , (lowered), with brief stochastic pulses (modeling thalamocortical relay rebound during spindles). The closure rules produce:

Band	Computation	Amplitude
Delta		
Sigma		
Theta		
Alpha		(→ 0 during pulses)
Beta		
Gamma		(transient with GLU bursts)

The profile is dominated by delta and the spindle band (sigma, prominent during the GLU phasic pulses) — the canonical electrophysiological signature of NREM sleep with spindles riding on slow oscillations (Steriade, 2006). Alpha is now low (), correctly collapsed relative to its waking value of (§4.4.1): the conjunctive closure gates alpha on noradrenergic arousal, and the NREM-analog NE collapse that frees the spindle band simultaneously removes alpha — so the framework no longer mis-reports a prominent NREM alpha, as a tonic-GABA-only rule would (GABA is *higher* here than in waking). Beta is suppressed to near-zero by the collapsed NE-tonic, in agreement with the well-documented loss of vigilance-band activity in NREM. The framework’s CFC envelope (eq. 35) consequently reaches high values when GLU phasic bursts coincide with high delta amplitude — the framework’s mechanistic correlate of slow-oscillation-spindle coupling, which has been independently identified as the substrate of NREM-dependent memory consolidation (Möller & Born, 2011).

Critically, the framework does not require a sleep-flag or a wake-flag to produce this signature. The transition from the wake regime of §4.4.1 to the NREM-analog regime of

§4.4.3 is purely a matter of shifting the tonic neurotransmitter mixture, and the band signature follows automatically from the closure. This is a non-trivial design property: explicit closure rules over a sufficiently rich NT vector are expressive enough to capture qualitatively distinct brain-state regimes without external regime labels.

4.4.4 Summary

The three regimes of this section establish that the framework's closure rules generate band signatures qualitatively consistent with the empirical EEG correlates of the corresponding brain states. The signatures are not arbitrary: they fall out of the closure matrix and the per-band biological motivations of §3.5.2, and they vary smoothly with the underlying neurotransmitter state. The framework therefore provides a closed-form mapping from neurochemical configuration to expected oscillatory signature, which is the central operational claim of Section 3.5. We turn next to the cross-frequency coupling envelopes, which combine multiple band amplitudes into the framework's signatures of associative encoding, attentional control, and consolidation.

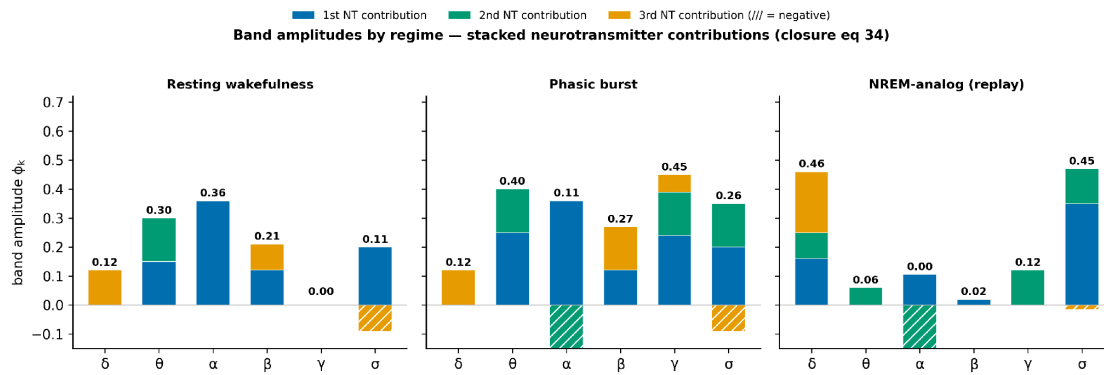


Figure 4.4: Band amplitudes by regime

Figure 4.4. Band amplitudes by cognitive/brain-state regime — stacked neurotransmitter contributions (closure eq. 32). Three representative regimes from §4.4. Each bar shows the total band amplitude with stacked contributions from the neurotransmitters in that band's closure rule (two or three terms per band; hatched bars below zero indicate negative contributions); the alpha band is the conjunctive GABA·NE exception (§3.5.2) and is shown as a single bar. Annotated values are the clamped total. *Resting wakefulness*: alpha (α) leads, with theta (θ) just below and beta (β) lower; gamma (γ) is zero in the absence of phasic drive. *Phasic burst*: gamma (γ) rises sharply from the DA/GLU/ACh phasic burst; beta and sigma also increase. *NREM-analog*: delta (δ) and sigma (σ) dominate; alpha (α) is low (α) and beta (β) floored (β), both via the collapsed NE-tonic. Regime vectors as specified in §4.4; band closure weights from DEFAULT_BAND_DERIVATION_RULES.

4.5 Cross-frequency coupling under task-relevant states

The three cross-frequency coupling envelopes defined in §3.5.3 are derived quantities (products of band amplitudes) but their interpretive value lies in their *state-specificity*:

each envelope reaches substantial magnitude only under a distinct combination of tonic and phasic neurotransmitter activity, and the resulting pattern of which CFC envelope dominates in which state constitutes a structural signature of brain state — one that depends on the sign pattern of the closure rules rather than on their hand-tuned magnitudes. In this section we walk through the three envelopes under their canonical functional regimes (associative encoding, inhibitory control, and NREM consolidation) and conclude with a contrastive prediction: that these three active regimes are each dominated by a different envelope and are jointly distinguishable against a quiescent resting baseline, in which no envelope is substantially engaged. As we make precise in §4.5.4, this is a claim about three active coupling regimes plus a baseline, not four orthogonal signatures.

4.5.1 Theta–gamma coupling during associative encoding

Theta–gamma coupling has been identified as the substrate of temporal binding and associative memory formation, with phase-amplitude coupling of gamma to theta phases providing a neural code for sequential ordering within episodes (Lisman & Jensen, 2013). The framework predicts the envelope should rise from near-zero in resting states to substantial magnitude during episodes in which both theta-supporting and gamma-supporting neurotransmitter conditions are simultaneously present.

We consider an associative encoding event in a socially salient, reward-associated context: $\langle \text{OXT}, \text{5-HT} \rangle$, with $\langle \text{ACh}, \text{DA} \rangle$. Here theta is supported by the cholinergic (ACh) and dopaminergic (DA) terms of its closure rule, and gamma by the phasic dopaminergic, glutamatergic, and cholinergic drive; the elevated OXT and 5-HT index the social/affective context of the encoding event but do not themselves enter any band closure under the present rules — see Appendix C. Applying the closure rules:

The envelope is substantially elevated from the resting baseline (where $\gamma = 0$, so $\theta = 0$). The framework therefore predicts that the theta–gamma envelope is a transient signature of associative encoding events specifically, with magnitude proportional to both the salience of the encoded stimulus (driving phasic glutamate, dopamine, and acetylcholine) and the cholinergic and dopaminergic tone that supports theta. This is consistent with the empirical finding that theta–gamma coupling in the hippocampus and prefrontal cortex correlates with successful encoding and depends on cholinergic drive (Lisman & Jensen, 2013; Canolty & Knight, 2010).

4.5.2 Alpha–beta coupling during impulse suppression

Alpha–beta coupling has been characterized as a signature of inhibitory control: alpha indexes thalamocortical inhibitory gating, beta indexes attentional precision and the maintenance of current task sets, and their joint elevation occurs during sustained

effortful suppression of distractor responses (Engel & Fries, 2010; Klimesch, 2012). The framework predicts substantial during regimes that combine elevated inhibitory tone with heightened vigilance.

A representative inhibitory-control configuration is: and (jointly supporting alpha through the conjunctive closure, with NE also driving beta), , (supporting beta), and no phasic glutamate; and index the broader affective/physiological context but no longer enter the alpha closure (§3.5.2). The closure rules produce:

At , this envelope reaches a substantial value under inhibitory control, well above the resting baseline and engaged only under its respective regime configuration. The alpha–beta envelope is near-zero during encoding (where gamma-supporting phasic activity dominates and the alpha and beta contributions are not jointly elevated) and near-zero during NREM (where collapsed NE-tonic drives both beta and the conjunctive alpha term to floor). It is, however, also the largest envelope in quiet resting wakefulness, but at much lower amplitude (≈ 0.08 ; see §4.5.4), so the alpha–beta envelope does not by itself separate resting from inhibitory control. What identifies the inhibitory-control regime is the *conjunction* of a dominant alpha–beta envelope with substantial amplitude; quiet rest shares the former but not the latter.

4.5.3 Delta–sigma coupling during NREM consolidation

The third envelope was partially developed in §4.4.3, where we noted that the NREM-analog tonic mixture produces high delta and intermittently high sigma (when glutamatergic phasic pulses drive thalamocortical relay rebound). Here we compute the explicit envelope. Using the NREM configuration of §4.4.3 with during a replay event:

During inter-replay quiescent periods (no GLU phasic), drops to , and the envelope to . The framework therefore predicts a *pulsing* delta–sigma envelope during NREM, with peaks at glutamate-replay events and elevated-but-lower baseline between them. This pattern matches the empirical signature of slow-oscillation–spindle coupling during NREM consolidation, in which spindle peaks ride on the upstates of slow oscillations and the coupling moments are the locus of hippocampal-cortical memory transfer (Steriade, 2006; Mölle & Born, 2011).

4.5.4 Contrastive prediction: state-specific CFC signatures

Together, the three active regimes plus a baseline establish the contrastive structure the framework predicts. We computed the three CFC envelopes for each regime using the

framework's actual closure weights (Appendix C) on the regime configurations of §4.4–§4.5; the qualitative levels below are the prediction, with verified computed magnitudes in parentheses:

Regime				Character
Resting wakefulness	NEGLIGIBLE (0.00)	LOW (0.08)	NEGLIGIBLE (0.01)	quiescent baseline — no envelope substantially engaged
Associative encoding	ELEVATED (0.10)	NEGLIGIBLE (0.00)	NEGLIGIBLE (0.00)	$\theta\gamma$ -dominated
Inhibitory control	NEGLIGIBLE (0.00)	ELEVATED (0.37)	LOW (0.04)	$\alpha\beta$ -dominated
NREM consolidation	NEGLIGIBLE (0.01)	NEGLIGIBLE (0.00)	HIGH (0.21)	$\delta\sigma$ -dominated

The structure here is *three distinguishable active coupling regimes* — encoding on $\theta\gamma$, controlled wakefulness on $\alpha\beta$, NREM on $\delta\sigma$ — read against a *quiescent resting baseline* in which no envelope is substantially engaged. It is important to state this accurately rather than as a clean four-way diagonal: resting wakefulness and inhibitory control both have their largest envelope on $\alpha\beta$, and the framework does *not* claim a distinct envelope for each of the four states. What distinguishes resting from inhibitory control is not *which* envelope dominates but its *amplitude*: $\alpha\beta$ sits low (0.08) in quiet rest and rises substantially (0.37) under inhibitory control, because the conjunctive alpha drive (high in the controlled-wakefulness regime, where both GABAergic inhibitory tone and noradrenergic arousal are elevated) and the noradrenergic beta drive both increase together. The prediction is therefore that the three active regimes are mutually distinguishable by *which* envelope dominates, and that the quiescent baseline is distinguishable from inhibitory control by the *amplitude* of $\alpha\beta$.

This contrastive pattern is the framework's central structural prediction about brain state, and it is important to be precise about what is and is not being predicted. The specific envelope magnitudes (0.10, 0.37, 0.21, and so on) are *not* the prediction — they are arithmetic consequences of closure weights we chose by hand. What the framework predicts is the *structure*: that the three active regimes are each dominated by a different envelope (encoding→ $\theta\gamma$, controlled wakefulness→ $\alpha\beta$, NREM→ $\delta\sigma$), that this assignment follows the sign logic of the closure rules rather than their exact coefficients, and that the quiescent baseline sits below all three with no envelope substantially engaged. This structural claim does not depend on the hand-tuned constants (it survives any choice of closure weights that preserves the sign pattern of the NT-to-band contributions) and it is the claim a sufficiently long EEG recording with concurrent task labels could falsify: if the three active regimes did *not* segregate by dominant envelope in the predicted assignment, the framework would be wrong in a way no recalibration

could repair. To be careful about the claim, we distinguish it from three nearby bodies of work. First, the empirical state-specificity of CFC is well established — theta–gamma in awake encoding states (Lisman & Jensen, 2013; Canolty & Knight, 2010), slow-oscillation–spindle (here delta–sigma) coupling in NREM (Mölle & Born, 2011; Steriade, 2006) — and we do not claim it as a discovery; it is the empirical backdrop. Second, the *conceptual* claim that neuromodulator balance determines which rhythm dominates in which state has been made before, most directly in the monoamine–acetylcholine balance hypothesis (Vakalopoulos, 2014), which argues that alpha rhythm reflects relative cholinergic predominance and delta rhythm relative monoaminergic predominance, with sleep-stage EEG signatures tracking relative neuromodulator levels. That is, qualitatively, a verbal statement of the present framework’s organizing idea; our contribution relative to it is to replace the verbal balance argument with an explicit closed-form generative map that produces a falsifiable contrastive envelope table from a specified concentration vector. Third, and most directly, generative neuromodulator-driven models of band-by-state activity already exist: thalamocortical neural-mass and biophysical models induce waking/NREM/REM transitions and stage-specific spectra (including spindles and the delta–spindle relationship) by varying parameters tied to the action of acetylcholine, noradrenaline, and serotonin (Schellenberger Costa et al., 2016; Krishnan et al., 2016). These are genuinely generative and genuinely neuromodulator-driven, so the unqualified claim that no such model exists would be false.

What none of these does is the framework’s specific construction: a single-node, closed-form algebraic map from an explicit twelve-system neurotransmitter-and-receptor-state vector to a contrastive cross-frequency-envelope-by-state table, via a sign-structured concentration→band closure, with no intervening population-dynamics simulation. The thalamocortical models route neuromodulation in as a few lumped biophysical parameters (a conductance, a gain, a noise level) and generate rhythms through simulated population dynamics; the balance hypothesis is verbal; the empirical CFC literature measures rather than derives. The residual that survives all three (closed-form, explicit multi-NT vector, sign-structured closure, contrastive envelope table, robust to its weakest sign entries above) is narrower than a claim of inventing chemistry-driven oscillation modeling, but it is the claim the framework can actually defend. We state it in those terms deliberately.

This contrastive prediction is the framework’s most specific operational claim about brain dynamics; we return to its empirical testability in Section 6 (Limitations and future work). Because the prediction rests entirely on the sign pattern of the closure rules, its robustness is the robustness of those sign assignments, which we enumerate and grade entry by entry in Appendix C. The theta closure rests on its two best-supported drivers (septal cholinergic input and dopaminergic gating); the alpha closure is a conjunctive GABA×NE term (GABAergic inhibitory tone gated by noradrenergic waking arousal; §3.5.2), reflecting its status as the relaxed-wakefulness rhythm. Every closure entry is SOLID or SUPPORTED by a primary citation; the only two candidate terms that

didn't clear that bar (the serotonergic and cannabinoid contributions originally considered for alpha) were excluded from the rule entirely rather than kept at a weaker grade (Appendix C). Across all four regimes, the dominant envelope is encoding→ $\theta\gamma$ (0.10), inhibitory control→ $\alpha\beta$ (0.37), and NREM→ $\delta\sigma$ (0.21), carried entirely by the best-supported entries (the glutamatergic, cholinergic, and dopaminergic drive of gamma and theta, the conjunctive GABA×NE basis of alpha, the GABAergic basis of the spindle band, the noradrenergic and cholinergic drive of beta, and the noradrenergic suppression of the spindle band) and independent of any contested sign assignment. This is the appropriate standard for the claim: a structural prediction built entirely on its best-supported inputs.

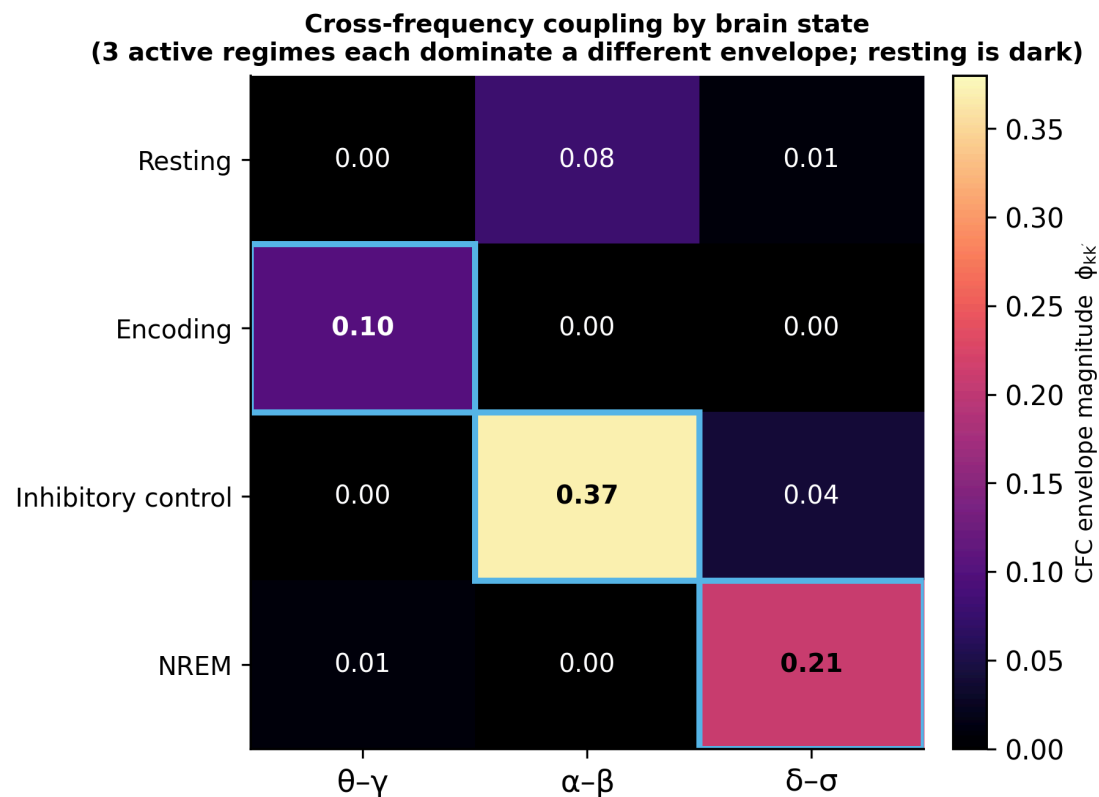


Figure 4.5: Cross-frequency coupling by brain state

Figure 4.5. *Cross-frequency coupling envelopes by brain state.* grid: rows are regimes (resting wakefulness, encoding, inhibitory control, NREM), columns are CFC envelopes (–, –, –). Cell color encodes envelope magnitude (magma colormap, range); bold values with sky-blue outlines mark the dominant envelope of each active regime. The contrastive prediction of §4.5.4: three active regimes each dominate a different envelope (encoding ; inhibitory control ; NREM) against a quiescent resting baseline (all envelopes). The resting-vs-inhibitory-control distinction is a magnitude contrast within the – column (0.08 vs. 0.37), not a separate column — the framework predicts three active coupling regimes plus a quiescent baseline, not four orthogonal signatures.

Values from the §4.5.4 table; all twelve cells computed from the closure rules of Appendix C.

4.6 Predictions under receptor and neurotransmitter manipulations

The framework's principal value as a hypothesis-generating tool comes from its ability to characterize the multi-system consequences of specific receptor and transporter manipulations. Because the framework couples 12 neurotransmitter systems through receptor pharmacodynamics and oscillatory closure, a perturbation applied to a single receptor population propagates through both regulatory and oscillatory channels.

It is important to separate two kinds of output the framework produces under manipulation, because they carry very different evidential weight. The first kind is a *consistency check*: a manipulation whose primary effect is already established textbook pharmacology, where the framework's value is only that it does not embarrass itself — it reproduces the known result through its own machinery rather than contradicting it. The second kind is a *novel structural prediction*: a multi-channel or cross-system signature that the field has not characterized and that single-system models cannot produce, which is where the framework's actual scientific contribution lies. We are explicit about which is which in each case below, because conflating them would let a sanity check masquerade as a discovery. The retrodictions (§4.6.1 first part, §4.6.3 concentration effects) establish that the framework is grounded; the novel predictions (the oscillatory side-signatures, the contrastive CFC table of §4.5.4, the chronic-versus-acute multi-channel distinction, and the 5-HT_{2A} amplitude/coupling dissociation) are the contribution.

We work through four representative manipulations, each motivated by a clinically or pharmacologically familiar intervention.

4.6.1 Chronic D₂ antagonism: receptor upregulation cascade

The D₂ receptor is the primary inhibitory dopamine autoreceptor and is the principal target of classical antipsychotic drugs. We model chronic D₂ antagonism as a sustained suppression of effective signaling from D₂ receptors, implemented in the framework by holding the receptor in a state of low binding regardless of ligand presence (functionally equivalent to a continuously occupied receptor whose downstream coupling is severed). Two predicted consequences follow.

First, as a consistency check: sustained low effective signaling from D₂ receptors satisfies the upregulation condition of eq. (28), so the receptor population transitions to UPRREGULATED, with density boost, sensitivity boost, and gate elevation to, giving the compound supersensitivity factor of (§4.3.2) — textbook pharmacology documented for nearly five decades and clinically associated with tardive dyskinesia upon dose reduction (Pierce et al., 2002; Stahl, 2013), reproduced here through the framework's own regulatory machinery rather than asserted.

Second, and this is the genuinely novel part: the framework predicts an *oscillatory* side-signature that the upregulation story alone does not anticipate. Through the D2-tonic contribution to the theta closure rule (§3.5.2), removing D2-mediated autoreceptor feedback disinhibits tonic DA release, raising $\bar{c}$ and therefore $\bar{c}_\theta$. The framework thus predicts a measurable theta elevation under chronic D2 antagonism, on a different timescale from the upregulation cascade (oscillatory shift over hours; upregulation over days–weeks). This is a risky, cross-channel prediction: it links a receptor manipulation to a band-amplitude shift through the closure table, and it could straightforwardly fail. The two effects are jointly testable (chronic antipsychotic treatment should produce elevated theta acutely and D2 upregulation chronically) and it is the theta limb, not the upregulation limb, that constitutes the framework’s claim.

4.6.2 5-HT2A receptor blockade: a receptor-layer, not closure-layer, effect

5-HT2A antagonism, a component of the pharmacology of atypical antipsychotics and a primary target of psychedelic-blocking agents, is modeled in the framework as a per-receptor attenuation of effective signaling at the 5-HT2A subtype without affecting 5-HT-tonic levels. Under the present closure (§3.5.2, Appendix C), the alpha-band amplitude carries no serotonergic term at all (the alpha closure rests on the conjunctive term and a phasic-glutamate desynchronization term; §3.5.2), so 5-HT2A blockade (and serotonergic state generally) has no closure-level effect on $\bar{c}_\alpha$ at any concentration. This is not a dissociation the model engineers; it is the model declining to assign a band-amplitude role to a relationship the literature does not support at the concentration level. The empirically documented effects of 5-HT2A activity on cortical processing — pyramidal-cell excitability, desynchronization, and the alpha-power suppression reported under psychedelic 5-HT2A agonism (Kometer et al., 2013; Muthukumaraswamy et al., 2013) — are receptor-occupancy phenomena, consistent with their absence from a tonic-concentration-driven closure rule. The framework does not currently model a receptor-occupancy-to-band-amplitude pathway for any subtype; extending the closure mechanism to admit such a pathway, rather than only concentration-driven terms, is a natural direction for future work and is noted in Section 6.

4.6.3 Selective reuptake inhibition: OU baseline shift

Selective transporter inhibition (for example, methylphenidate [DAT inhibition] or selective serotonin reuptake inhibitors [SERT inhibition]) is modeled in the framework as a reduction in the transmitter-specific reuptake coefficient $\bar{c}_r$ in eq. (6). With reduced clearance, the OU stationary mean of the tonic component shifts upward. From the analytical form of eq. (4) at steady state:

A reduction in $\bar{c}_r$ produces an unambiguous upward shift in the steady-state concentration, in proportion to the relative magnitude of the reuptake term within the denominator. For dopamine, where reuptake is the dominant clearance channel, this gives elevated steady-state DA under DAT inhibition; for serotonin, the same under SERT inhibition — a consistency check rather than a prediction, since the result is both

known empirically and close to true by construction of eq. (57); the framework would be in trouble if it did *not* reproduce it.

The novel content is again at the oscillatory layer, and the closure of Appendix C makes the prediction sharp rather than diffuse. Elevated θ raises θ through the theta closure rule. Elevated α , by contrast, raises no band amplitude at all: serotonin appears in none of the six closure rules (Appendix C), since no candidate serotonergic term in the theta or alpha rules survives review against the electrophysiology literature. The framework therefore predicts an *asymmetric* signature between the two interventions: DAT inhibition should produce a measurable theta-amplitude elevation through the closure pathway, while SERT inhibition, despite raising synaptic 5-HT by the same OU mechanism (eq. 57), should produce no corresponding closure-level band-amplitude signature. This is a sharper and more falsifiable claim than the symmetric “both raise oscillatory power, just in different bands” version it replaces — it predicts an outright absence of one of the two effects, not merely a difference in profile, and it is correspondingly easier for EEG data to contradict. We note explicitly that this null is a property of which serotonergic effects survived literature verification in the present closure table, not a claim that no serotonergic oscillatory effect exists; SSRIs are known to alter EEG spectral power (see, e.g., the tonic/SSRI literature reviewed in Appendix C’s alpha entries), and a closure mechanism richer than tonic-concentration-linear terms (for instance, one admitting receptor-occupancy or frequency-dependent terms) might recover such an effect. We return to this gap in Section 6.

4.6.4 Chronic stress: multi-band, multi-mechanism signature

Chronic stress is modeled in the framework through sustained elevation of cortisol, with downstream effects on both receptor regulation (§3.4.6 emotion-modulated plasticity) and the beta-band closure rule (where β has weight w_β). The framework predicts a multi-component signature: sustained moderate elevation of θ through the closure, sustained per-receptor sensitivity adjustments through the plasticity layer, and slow accumulation of receptor adaptation cascades for chronically stimulated stress-axis receptors (CRH, glucocorticoid). The signature is multi-system and multi-timescale by construction — the framework’s principal claim is that chronic stress does not reduce to a single-system perturbation but produces coordinated changes across at least three independently observable channels (band amplitude, receptor sensitivity distribution, regulatory state cycling).

This kind of multi-component prediction is what the framework adds beyond single-system models. A model of cortisol effects on one receptor population yields a single-system prediction; a model of cortisol effects on the beta-band closure yields an oscillatory prediction; the present framework yields both, plus the receptor-regulatory coupling between them, plus the cross-frequency coupling consequences that follow if the elevated θ shifts the framework’s θ envelope.

4.6.5 The shape of testable claims

The four manipulations are representative rather than exhaustive. The consistency checks among them (D2 antagonism causing upregulation, reuptake inhibition raising concentration) are necessary but are not the contribution. The genuine predictions are the cross-channel signatures the field has not characterized: that single-receptor perturbations produce *coordinated* multi-channel responses (concentration, regulatory state, and oscillatory band) that single-system models cannot generate; that two interventions sharing a coarse mechanism (DAT vs SERT inhibition) are predicted to be *asymmetric* at the oscillatory level, with one producing a closure-level theta signature and the other producing none; that 5-HT_{2A} receptor blockade is correctly modeled as a receptor-layer rather than closure-layer effect, with no consequence for alpha-band amplitude; and, most distinctively, the contrastive CFC segregation of §4.5.4. The common structure of these genuine predictions is that a perturbation propagates through the coupled channels and produces a signature whose *sign pattern and cross-channel coordination*, not its hand-tuned magnitudes, are the falsifiable content. This is what the framework adds beyond single-system models, and it is where the scientific risk, and therefore the scientific value, is concentrated. We discuss the implications for empirical work in Section 5.

The model behavior described across Sections 4.1–4.6 establishes the framework's structural output claims — the signs, directions, orderings, and timescale separations that follow from its design and survive the specific choice of constants. The next section places these claims in the broader context of prior computational neuroscience work and discusses their methodological and conceptual implications.

5. Discussion

The preceding sections developed a mathematical framework for coupled twelve-system neurochemical dynamics, established its qualitative and quantitative behavior under canonical input regimes, and worked through specific predictions for receptor and neurotransmitter manipulations. In this section we step back from the technical material and place the framework in the context of prior computational neuroscience work, articulate the principle that organizes its design choices, examine its biological plausibility, and offer a focused methodological note on the one element of its machinery adapted from quantitative finance.

5.1 Relationship to prior computational modeling of neuromodulators

The framework is best understood as an extension and integration of the dopamine-and-serotonin computational lineage that has dominated computational neuroscience modeling of neuromodulators since the late 1990s. Within that lineage, three threads of work have been particularly productive: the dopamine-as-reward-prediction-error tradition that originates with Schultz et al. (1997) and is mathematically formalized in temporal-difference reinforcement learning (Sutton

& Barto, 2018); the dopamine-serotonin interaction modeling of Cools (2008) and Boureau & Dayan (2011); and the broader multi-neuromodulator metalearning theory of Doya (2002), which proposed that four neuromodulators (DA, 5-HT, NE, ACh) jointly index four distinct learning parameters (value, exploration-exploitation balance, prediction salience, attentional gain). A parallel thread has addressed receptor-subtype dynamics specifically within the basal ganglia, where the distinct roles of D1- and D2-expressing striatal medium spiny neurons in approach and avoidance learning have been formalized in computational models (Frank, Seeberger, & O'Reilly, 2004; Frank, 2005). A further thread has applied computational modeling to clinical phenotypes through the lens of computational psychiatry (Daw, Niv, & Dayan, 2005; Dayan & Huys, 2009).

The present framework extends this lineage along three orthogonal axes simultaneously. First, it generalizes from the canonical two- or four-neuromodulator setting to a full twelve-system coupling, providing a substrate for the under-modeled neuromodulators (oxytocin, the endocannabinoid system, opioids, histamine, cortisol and CRH) that play documented roles in motivated cognition but rarely appear in computational neuroscience models. Second, it adds an explicit receptor regulatory layer through the state machine of Section 3.4 — a layer that the dominant computational neuroscience lineage typically treats as an implicit parameter rather than a dynamical variable. Third, it adds the oscillatory closure of Section 3.5, which positions brain rhythms as an integral part of the dynamics rather than as an epiphenomenal output to be predicted post hoc. Each of these extensions is independently defensible against the prior work, but their joint inclusion is what enables the framework's most distinctive predictions, for example the contrastive CFC signature across functional states (§4.5.4) and the multi-channel manipulation signatures (§4.6.5), which neither pure neurotransmitter models nor pure oscillatory models can produce.

We do not view the framework as a replacement for the dopamine-serotonin computational lineage; the more specialized models within that tradition retain their predictive power for the specific decision-making and learning paradigms they were designed to address. The present framework operates at a different level of abstraction, providing a *substrate* on which more specialized cognitive models can be built rather than competing with them at the cognitive-level computation. Concretely: a reinforcement-learning model that requires a dopamine-RPE signal can read out from the framework; a model of selective serotonin reuptake inhibition can apply the framework's eq. (57); a model of receptor regulatory dynamics in chronic antipsychotic exposure can read out the cascade trajectory of Section 4.3. The framework adds layers below the level at which most existing computational neuroscience models operate.

5.2 The multi-system coupling principle

The framework's defining design choice is to couple twelve neurotransmitter systems through both receptor pharmacodynamics and oscillatory closure. This choice has costs

(twelve-system coupling adds parameters, increases the surface area for empirical calibration, and complicates the analysis of single-perturbation experiments) but it is the source of the framework's distinctive predictions. We summarize the principle here.

Single-system or paired-system neuromodulator models can produce a wide variety of single-system behaviors, but they cannot produce the coordinated multi-channel signatures that distinguish chronic from acute perturbations, that distinguish one brain state from another by closure signature rather than state label, or that produce three-mechanism habituation as a *system-level* property (§4.2.2) rather than a pasted-together combination of unrelated effects. These behaviors emerge only when the architecture provides simultaneously the multi-NT substrate, the receptor regulatory layer, and the oscillatory closure. The framework's contribution at this conceptual level is therefore the demonstration that such an architecture is mathematically tractable, biologically motivated, and capable of producing recognizable empirical signatures — not the demonstration that any single coupling is correct.

A productive way to read the framework is therefore as an architecture for *parsimonious* multi-system prediction: it commits the modeler to a small number of design choices (the closure matrix entries, the state machine thresholds, the modulation coefficients) and then derives the full multi-channel response. Several predictions developed in Section 4 (the three-mechanism habituation, the multi-cascade attenuation, the closure-loop feedback amplification, the contrastive CFC signature, the multi-channel manipulation signature) are each *structural* predictions: claims about the sign, the direction, the relative ordering, and the timescale separation of the response, rather than about the specific magnitudes our hand-tuned constants produce. So framed, each is directly testable against the empirical literature in a way that does not collapse into recalibration, because the structural features survive any choice of constants that preserves the sign logic of the design. This distinction matters because the framework's specific numerical values are consistency-calibrated rather than empirically fit (§5.3), and a prediction that depended on those values would not be a genuine prediction at all.

5.3 Biological plausibility and parameter grounding

The framework's parameters fall into three categories with respect to empirical grounding. The first category, comprising the structural choices (the four-state receptor machine, the multiplicative effective-signaling proxy, the OU drift form, the multiplicative square-root noise), is grounded directly in the pharmacology literature: each structural choice has a clear precedent in receptor biology (Pierce, Premont, & Lefkowitz, 2002; Gainetdinov et al., 2004; Hanyaloglu & von Zastrow, 2008), in operational pharmacology (Black & Leff, 1983; Rang et al., 2019), or in biophysical noise theory (the Poisson character of vesicular release; the CIR mathematical lineage). The second category, comprising the *order-of-magnitude* parameter ranges, is similarly grounded: receptor dissociation constants span the empirically documented two-order-of-magnitude range; desensitization timescales fall in the seconds-to-minutes range; internalization timescales fall in the minutes-to-hours range; upregulation timescales fall in the

hours-to-days range. These ranges match the relevant biological literature (Pierce et al., 2002; Gainetdinov et al., 2004).

The third category, comprising the *specific numerical values* of weights in the closure matrix, coefficients in the modulation rules, and thresholds in the state machine, is less directly empirically grounded. These values were chosen to produce qualitatively reasonable behavior under the canonical regimes of Section 4 and are calibrated for internal consistency rather than fit to specific empirical recordings. Two consequences follow. First, individual predictions of Section 4 should be interpreted as predictions of the *shape* of the framework's response rather than precise numerical fits; a successful framework-prediction is one where the predicted signature is qualitatively present in empirical data, not one where a specific magnitude is recovered. Second, the framework should be understood as hypothesis-generating rather than empirically validated: it provides a principled architecture for organizing predictions, but the predictions themselves require empirical work to evaluate.

We view this as a deliberate trade-off. A framework that committed to specific empirically-fit values for every coefficient would be a calibrated model of a specific dataset; a framework that left every coefficient free would have no testable content. The present framework occupies the middle ground: structural choices and order-of-magnitude calibrations grounded in literature, specific coefficients chosen for internal consistency and open to refinement by future empirical work. This middle ground is appropriate for the framework's primary use case as a substrate for downstream cognitive models, where consumers will typically reconfigure the per-receptor parameters for their specific application rather than rely on the defaults.

5.4 Methodological note: positivity-preserving noise from quantitative finance

As detailed in §2.2, only one piece of the framework's mathematical machinery is a genuine cross-disciplinary import: the CIR-style square-root diffusion (Cox, Ingersoll, & Ross, 1985) used for the multiplicative noise term of eq. (7). The OU drift and the Euler–Maruyama integration scheme are native to computational neuroscience and numerical analysis respectively, so neither is an import in any meaningful sense.

The methodological point, then, is narrower and more useful than a blanket “imported from finance” framing: positivity-preserving square-root diffusion is a natural fit for any bounded, nonnegative biological concentration variable whose release variability scales with rate, it solves the boundary-handling problem that additive noise creates (§3.2.5), and — given that additive Gaussian noise remains the default in neural SDE modeling — it deserves wider use at the concentration level than it currently gets. A companion methodology paper develops the broader SDE–CTMC architecture as a modeling approach independent of the specific neurochemical application.

5.5 Implications and open directions

The framework's most consequential implication, on our reading, is methodological. Multi-system coupling is mathematically tractable when the right tools are imported, biologically motivated when the structural choices are grounded in pharmacology, and capable of producing recognizable empirical signatures when the closure rules are explicit. None of these properties was obvious in advance. Each is established by the framework's actually working: the closure rules of §3.5.2 do produce qualitatively right band signatures (§4.4), the regulatory cascade of §3.4 does produce the expected timescale separation (§4.3), and the manipulation predictions of §4.6 do reproduce known clinical phenomenology (D2 upregulation under chronic antagonism, OU baseline shift under reuptake inhibition).

Open directions are discussed in §6 below; here we note only that the framework is most valuable as an infrastructure on which other groups can build. The 12-system coupling, the receptor state machine, and the oscillatory closure are independent modules in the framework's implementation; researchers can adopt any subset of the framework's machinery for their own modeling work without committing to the full architecture. We view this modularity as part of the framework's contribution.

6. Limitations and future work

The framework presented in this paper is best understood as a starting point rather than a finished model. We have argued throughout for its biological grounding and methodological coherence, but a number of limitations are worth surfacing — both because they shape how the framework should be used in its present form and because they identify the most productive directions for future development. We organize the limitations into three categories (scope, calibration, and implementation) followed by a brief discussion of future empirical and theoretical work.

6.1 Limitations of scope

The framework is a *single-compartment* model. Each receptor's effective concentration is computed via the localization parameter of eq. (19), which is a useful coarsening of the spatial distinction between synaptic, perisynaptic, and extrasynaptic populations, but does not model spatial diffusion or volume transmission in any geometric sense. A spatially extended version of the framework (in which neurotransmitter concentrations evolve in a 2D or 3D compartmental mesh and receptors are placed at specific positions relative to release sites) would provide a richer substrate for predictions about volume transmission and would extend the framework's reach into spatially-resolved electrophysiology data. We see this extension as the single most consequential future development for the framework, and we have deliberately preserved the architecture's modularity so that a spatial layer can be added without disturbing the existing kinetic and regulatory machinery.

The oscillatory layer is *amplitude-only* in the framework's primary formulation. Band amplitudes are tracked and back-modulate kinetic parameters; band phases are tracked as state variables in the implementation but do not enter any of the modulation rules of §3.5.4. Phase-amplitude coupling experiments (for example, the explicit modeling of theta-phase-locked gamma bursts that the empirical CFC literature documents [Tort et al., 2010]) would require an extension of the modulation rules to include phase-dependent terms. The framework supports this extension structurally but does not implement it in the current form, and the predictions of §4.5 are correspondingly amplitude-envelope predictions rather than phase-resolved ones.

The framework applies a single slow regulatory state machine uniformly across receptor classes whose real signaling biology differs substantially in kind, as set out in §3.1. Fast ionotropic receptors (AMPA, NMDA, kainate, GABA-A) mediate millisecond-scale synaptic transmission that the four-state GPCR regulatory cycle does not represent; nuclear steroid receptors (the cortisol and CRH systems) act on genomic timescales of minutes to days through mechanisms entirely distinct from membrane-receptor desensitization. The framework's defense (that it models the slow neuromodulatory envelope common to all these systems rather than their divergent fast or genomic signaling) is a legitimate coarsening for the framework's intended purpose, but it is a real limitation on the framework's biological completeness. A more faithful version would equip the ionotropic populations with a fast-channel kinetic model and the steroid systems with an explicit genomic-delay model, gated by the ionotropic/metabotropic tagging the implementation already carries. We regard this differential-dynamics extension as one of the more important refinements for future versions, particularly for any application in which the fast transmission roles of glutamate and GABA, rather than their slow modulatory roles, are the object of study.

A related simplification is the omission of direct presynaptic heteroreceptor coupling, also noted in §3.1: the framework routes all inter-system interaction through shared receptor and oscillatory state rather than through direct concentration-to-concentration terms. Documented presynaptic NT–NT regulation is therefore captured only indirectly, and a version with explicit heteroreceptor terms is a candidate extension.

The framework does not specify the downstream cognitive architecture that consumes its effective signaling output. This is by design — the framework is intended as a substrate that can be paired with different cognitive architectures, and the per-receptor weight in downstream computation (the w that does not appear in the proxy of eq. 22) is left for the consuming architecture to specify. The cost of this design choice is that the framework cannot, on its own, produce *behavioral* predictions; behavioral predictions require pairing the framework with a cognitive model. We see this as appropriate for a substrate-level framework, but it should be made explicit that the framework is incomplete as a standalone tool for predicting cognition.

6.2 Limitations of calibration

As noted in §5.3, the framework's parameters partition into three categories with respect to empirical grounding: structural choices (well-grounded in pharmacology), order-of-magnitude parameter ranges (well-grounded in literature), and specific numerical values (consistency-calibrated rather than empirically fit). The third category includes the entries of the NT-band closure matrix in §3.5.2, the band-coefficient values in the modulation rules of §3.5.4, and many of the receptor regulatory thresholds in §3.4.3. These specific values produce qualitatively reasonable behavior across the regimes of Section 4, but they have not been fit to specific empirical datasets and are most appropriately interpreted as initial defaults open to refinement.

Four calibration issues deserve particular note. First, the theta- and alpha-band closures each omit candidate terms that point the wrong way against the electrophysiology literature. Oxytocin and serotonin are excluded from the theta closure: serotonin from the median raphe suppresses hippocampal theta (Vertes & Kocsis lineage) and oxytocin correlates negatively with theta power in vivo (Zhang, Karadas, et al., 2024); the theta closure instead rests on the canonical cholinergic generator (). Serotonin and the cannabinoid system are likewise excluded from the alpha closure: 5-HT_{2A} activation suppresses rather than enhances alpha (Komater et al., 2013), and CB₁'s documented oscillatory effects are on theta and gamma rather than alpha; the alpha closure instead takes the conjunctive form described below (§3.5.2, Appendix C), for the reasons given next. In both cases the contrastive CFC structure of §4.5.4 is preserved under these rules. One consequence, taken together, is that serotonin does not appear in any of the six band-closure rules: every serotonergic candidate term evaluated against the literature fails at either the sign or the existence level. We record this as an open gap rather than retaining an under-supported term for coverage's sake — the framework's existing predictions (§4.6.2, §4.6.3) are stated accordingly, including a prediction that SERT inhibition produces no closure-level oscillatory signature, which is itself falsifiable and may simply indicate that the present concentration-linear closure mechanism is the wrong tool for capturing serotonin's real (and well-documented) EEG effects, several of which appear to be receptor-occupancy- rather than concentration-driven. Extending the closure mechanism to admit non-concentration-linear terms is future work. A second issue is specific to the alpha band, and motivates the one non-linear term in the closure. Keying alpha to tonic GABA alone produces a theta- rather than alpha-led resting profile () and, worse, a *spuriously high* alpha in the NREM-analog regime, because tonic GABA is highest in slow-wave sleep, so a tonic-GABA rule is largest exactly where alpha should vanish. Both symptoms share one root cause: tonic GABA concentration is anti-correlated with wakefulness, whereas alpha is a waking rhythm. This is not repairable by re-weighting — raising the GABA coefficient lifts NREM alpha *more* than waking alpha, and raising resting GABA-tonic (the needed to lift alpha above theta) co-drives sigma to , placing spindle-band power in quiet wakefulness. The alpha closure is therefore conjunctive (§3.5.2): , encoding that alpha requires *both* inhibitory tone and intact noradrenergic waking arousal. With this single non-linear term the framework reproduces alpha-dominant eyes-closed rest (leading; §4.4.1) *and* correctly collapses

alpha in NREM (; §4.4.3), at no cost to the contrastive predictions (the resting-vs-inhibitory $\alpha\beta$ contrast in fact widens; §4.5.4). This is a deliberate departure from strict linearity, motivated by alpha's status as a state-gated rhythm rather than fitted to output: the *form* (a conjunction of inhibition and arousal) is biologically derived, while the scalar is a consistency calibration like the other closure weights. The residual limitation is that the noradrenergic dependence is monotonic, so the conjunctive term captures the sleeprelaxed-wake rising limb of alpha's inverted-U dependence on arousal but only partially captures its desynchronization at high arousal (via the term); a non-monotonic arousal dependence is left as future work. Second, the weakest-grounded entries that remain in the closure table are dose- or context-dependent rather than wrong-signed (gamma-DA, gamma-ACh, delta-MOR; Appendix C); these are flagged with their caveats in place rather than removed, since the direction is correct even where the magnitude is conditional. Third, the exposure-trace asymptote sits below the default internalization threshold, which we have framed as a deliberate property (§4.3.4) such that internalization is reached primarily through phasic-saturated regimes rather than tonic saturation alone. This framing is defensible biologically but represents a calibration choice that other groups may want to adjust. Fourth, the emotion-modulated plasticity table referenced in §3.4.6 is a per-emotion per-receptor configuration choice rather than an analytical derivation, and the framework's predictions hold for any choice of plasticity table satisfying basic monotonicity and boundedness conditions rather than for a specific calibrated table.

6.3 Limitations of implementation

The framework's current implementation, on which the worked examples of Section 4 are based, is a fully tested and reproducible system but is not feature-complete with respect to the framework specification.

The receptor state machine of §3.4 uses deterministic threshold-based transitions: the transition rates are step functions (eligible or ineligible) rather than smooth stochastic rate kernels, so transitions fire as soon as their accumulator conditions are satisfied. As argued in §3.4, we regard this as the appropriate level of description for the slow population-level regulatory envelope the framework models, not as a deficiency — a deterministic machine is analytically transparent, reproducible, and sufficient for every result in this paper. We flag it as a limitation only in the narrow sense that the framework does not, in this formulation, capture the trial-to-trial stochasticity of individual-receptor regulatory trajectories that a smooth-rate-kernel CTMC would. That stochastic generalization is a genuine extension rather than a correction, it is accommodated by the notation we already use, and it is developed in a companion paper; readers should not, however, read the present model as an incomplete approximation awaiting that paper, since the deterministic formulation is self-contained and adequate on its own terms.

The framework specifies modulation of receptor state transition timing by oscillatory band amplitudes (eq. 41), and the implementation includes a partial set of band-coefficient tables for specific receptor clusters; however, four documented

modulation rules for specific cluster–band pairs are not yet implemented (these are documented in supplementary materials as the framework’s outstanding implementation work). The omissions do not affect the qualitative behavior of the framework in the regimes of Section 4, but they limit the per-cluster differentiation of predictions and should be addressed before the framework is used for predictions specifically involving these cluster–band couplings.

The CTMC per-subtype differentiation is partially populated: the state machine of §3.4 operates per-receptor-subtype but several subtype-specific threshold tunings are currently set to default values rather than subtype-specific values. Predictions of Section 4 that depend on differential subtype behavior (in particular, several of the manipulation predictions in §4.6) should be interpreted as predictions of the *form* of the response rather than precise subtype-level differentiation.

Finally, a note on what the implementation’s test suite does and does not establish. The suite comprises 1,440 deterministic tests (the neurochem-specific subset; the prior figure counted the full cognitive-architecture test suite rather than this layer alone), and they verify implementation fidelity (that the code computes the equations of Sections 3 and 4 correctly and reproducibly under seeded RNG) rather than biological validity, which the tests cannot speak to and which we do not claim for them. We make this distinction explicit because a large test count can be misread as empirical support for the model; it is not. The tests establish that our equations are implemented as written, no more.

6.4 Future empirical and theoretical directions

Several directions for future work follow naturally from the limitations above and from the framework’s hypothesis-generating predictions.

Empirical calibration of closure weights. The most consequential calibration work would fit the NT-band closure matrix of §3.5.2 against simultaneous EEG/MEG and pharmacology data. Several modern paradigms permit such fitting in principle — for example, pharmacological manipulations with concurrent EEG, or natural-state recordings under PET-quantified receptor occupancy. We would expect such fitting to refine the closure weights without altering the framework’s qualitative predictions.

Empirical tests of manipulation predictions. The predictions of §4.6 — D2 antagonism producing both an upregulation cascade and a theta shift, 5-HT_{2A} blockade producing no closure-level alpha-amplitude change despite known receptor-layer effects on cortical processing, selective reuptake inhibition producing an asymmetric band signature (theta elevation under DAT inhibition, no closure-level signature under SERT inhibition), chronic stress producing multi-channel signatures — are all in principle testable in existing pharmacological-EEG paradigms. We see these as high-priority directions for empirical follow-up, and we note that a null result for the SERT-inhibition prediction specifically would be confirmatory of the framework rather than falsifying — it is the *positive* theta prediction under DAT inhibition, and the asymmetry between the two, that constitutes the falsifiable claim.

Theoretical extensions. The two extensions we view as most important are (i) a stochastic rate-kernel generalization of the receptor state machine (in which the deterministic transitions of §3.4 are replaced by smooth-rate-kernel transitions capturing individual-receptor variability) developed in the companion paper, and (ii) a spatially-extended formulation that adds a compartmental mesh and explicit volume-transmission dynamics. A third extension, also developed in the companion methodology paper, generalizes the SDE–CTMC architecture as a methodology for cognitive-process modeling beyond the specific neurochemical application.

Integration with cognitive architectures. The framework’s substrate-level character means that its full utility is realized only when paired with a cognitive architecture that consumes its outputs. Integrations with existing reinforcement-learning, decision-making, and cognitive-control architectures would test the framework’s claims about multi-system coupling at the behavioral level and would provide additional avenues for empirical validation.

We close by noting that the framework, despite the limitations enumerated above, provides a coherent and reproducible architecture for multi-system neurochemical modeling whose distinctive feature is the level of integration — twelve neuromodulator systems, a receptor regulatory layer, and a direct oscillatory closure combined in one coupled system. The individual components draw on established traditions, as Section 2 sets out; we are not aware of prior work combining them at this scope, though we make that as a statement about integration rather than about any single component’s novelty. The limitations identify directions for refinement rather than fundamental challenges to the framework’s coherence, and we hope they provide useful entry points for groups interested in building on this work.

Code and Data Availability

All code, configuration files, and tests are available at [DOI — to be assigned upon submission; planned Zenodo deposit]. The framework is implemented in Python 3 using `numpy.random.default_rng` for seeded pseudorandom number generation, ensuring byte-reproducible trajectories. The implementation includes 1,440 deterministic unit and integration tests that verify the code faithfully reproduces the equations described in this manuscript under seeded pseudorandom number generation; these establish implementation fidelity and reproducibility, not biological validity. Figures were generated by the figure-generation script included in the repository; all numerical results in Sections 4.1–4.5 (analytical predictions, ensemble statistics, band amplitudes, and CFC envelope values) are byte-reproducible under the random seeds reported in the text.

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Appendix E — Per-receptor coupling references, listed separately for now; to be merged into the main reference list above at submission.

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Appendix A — Per-transmitter kinetic parameters

Each of the twelve neurotransmitter systems is parameterized by nine kinetic constants governing its tonic/phasic stochastic dynamics (§3.2) and clearance. The values below are the framework’s default configuration (DEFAULT_NT_CONFIGS); consistent with the hypothesis-generating stance of §1 and §6, they are order-of-magnitude calibrations grounded in the relative kinetics of each system (e.g., fast cholinergic and glutamatergic phasic decay, slow peptidergic and steroid reversion), not values fitted to data. The final column names the biological clearance pathway represented by the `u_base/d_base` terms.

										Clear ance mech anism
NT										fat. rate
DA	0.50	0.10	1.00	0.05	0.10	0.10	0.05	0.020	0.001	DAT reupt ake + COM T/MA O
5-HT	0.55	0.05	0.50	0.03	0.06	0.08	0.03	0.015	0.001	SERT reupt ake + MAO

										Clear ance mech anism
NT									fat. rate	
NE	0.45	0.15	1.20	0.06	0.12	0.12	0.04	0.020	0.001	NET reupt ake + COM T/MA O
AC h	0.50	0.12	1.50	0.04	0.08	0.15	0.03	0.010	0.001	AChE hydrol ysis
OX T	0.40	0.03	0.30	0.02	0.04	0.05	0.02	0.010	0.001	Pepti dase
M OR	0.35	0.04	0.60	0.03	0.07	0.06	0.03	0.015	0.001	Pepti dase + diffusi on
CB 1	0.40	0.03	0.40	0.025	0.05	0.04	0.02	0.010	0.001	FAAH degra dation
Co rtis ol	0.30	0.02	0.20	0.015	0.03	0.03	0.015	0.010	0.001	11 β -H SD meta bolis m
CR H	0.25	0.08	0.80	0.04	0.09	0.10	0.04	0.020	0.001	Pepti dase
GA BA	0.60	0.10	1.00	0.04	0.08	0.12	0.04	0.020	0.001	GAT reupt ake + GAB A-T
GL U	0.55	0.12	1.30	0.05	0.10	0.15	0.05	0.020	0.001	EAAT reupt ake + Gln synth etase

NT										Clear ance mech anism
								fat. rate		
His ta mi ne	0.35	0.08	0.90	0.04	0.08	0.10	0.04	0.020	0.001	HNM T + MAO- B/DA O

Legend: = baseline concentration; = tonic OU mean-reversion rate; = phasic decay rate; , = tonic/phasic diffusion noise amplitudes (the per-NT coefficient of eq. 7); = reuptake coefficient; = enzymatic degradation rate; = diffusion/clearance rate; fat. rate = release-fatigue accumulation constant. Concentrations and amplitudes are dimensionless on ; rates are per integration tick.

Appendix B — Receptor configuration

The framework instantiates **30 receptor subtypes** across eleven of the twelve transmitter systems; cortisol is modeled as concentration-only, signaling through direct modulation with no entry in the receptor registry. Per-transmitter subtype counts range from one (OXT, MOR, CB1, CRH) to five (DA, 5-HT). Each subtype carries a dissociation constant (lower = higher affinity), an exposure-trace time constant that drives the regulatory state machine of §3.4, a signaling polarity, a per-subtype effective-signaling weight applied to its downstream contribution, and an ionotropic/metabotropic tag. Values are the default configuration (DEFAULT_RECEPTOR_CONFIGS merged with the per-family ReceptorSpec registry).

#	Receptor	Parent NT		Signaling		Weight	Ionotropic
1	DA_D1	DA	0.40	10.0	excitatory	1.00	No
2	DA_D2	DA	0.30	12.0	inhibitory	0.90	No
3	DA_D3	DA	0.20	15.0	inhibitory	0.80	No
4	DA_D4	DA	0.35	10.0	modulatory	0.85	No
5	DA_D5	DA	0.45	10.0	excitatory	0.95	No
6	5HT_1A	5-HT	0.30	15.0	inhibitory	1.00	No
7	5HT_1B	5-HT	0.35	12.0	inhibitory	0.85	No
8	5HT_2A	5-HT	0.40	10.0	excitatory	1.00	No
9	5HT_2C	5-HT	0.35	12.0	modulatory	0.90	No
10	5HT_3	5-HT	0.50	8.0	excitatory	0.80	Yes

#	Receptor	Parent NT			Signaling	Weight	Ionotropic
1 1	NE_alpha 1	NE	0.50	10.0	excitatory	1.00	No
1 2	NE_alpha 2	NE	0.25	12.0	inhibitory	0.90	No
1 3	NE_beta1	NE	0.40	10.0	excitatory	0.95	No
1 4	NE_beta2	NE	0.45	10.0	excitatory	0.85	No
1 5	ACh_nico tinic	ACh	0.50	8.0	excitatory	1.00	Yes
1 6	ACh_mus carinic	ACh	0.40	12.0	modulatory	0.90	No
1 7	OXTR	OXT	0.35	20.0	excitatory	1.00	No
1 8	MOR_mu	MOR	0.30	15.0	inhibitory	1.00	No
1 9	CB1	CB1	0.40	20.0	inhibitory	1.00	No
2 0	CRH_R1	CRH	0.45	10.0	excitatory	1.00	No
2 1	GABA_A	GABA	0.50	8.0	inhibitory	1.00	Yes
2 2	GABA_B	GABA	0.40	15.0	inhibitory	0.90	No
2 3	GLU_NM DA	GLU	0.50	10.0	excitatory	1.00	Yes
2 4	GLU_AM PA	GLU	0.60	6.0	excitatory	1.00	Yes
2 5	GLU_KAI NATE	GLU	0.55	8.0	excitatory	0.85	Yes
2 6	GLU_mGl uR	GLU	0.35	15.0	modulatory	0.90	No
2 7	HIST_H1	Histamine	0.45	10.0	excitatory	1.00	No
2 8	HIST_H2	Histamine	0.40	12.0	excitatory	0.80	No
2 9	HIST_H3	Histamine	0.25	15.0	inhibitory	0.60	No

#	Receptor	Parent NT			Signaling	Weight	Ionotropic
3	HIST_H4	Histamine	0.50	10.0	modulatory	0.40	No
0							

Legend: = dissociation constant; = exposure-trace time constant (drives the §3.4 state machine); Weight = per-subtype effective-signaling weight; Ionotropic = fast ligand-gated channel (Yes) vs. metabotropic/GPCR (No). The six ionotropic subtypes (5HT_3, ACh_nicotinic, GABA_A, GLU_NMDA, GLU_AMPA, GLU_KAINATE) run the same slow regulatory state machine as the metabotropic subtypes in the present formulation — the deliberate coarsening discussed in §3.4 and flagged for future differentiation in §6.

Appendix C — NT→band closure: per-entry sign justification

This appendix is the load-bearing evidence base for the framework’s structural predictions. Because the framework’s testable content rests on the sign pattern of the NT→band closure (§3.5.2), not on the hand-tuned magnitudes, every individual sign assignment must carry its own defensible primary citation. The table below enumerates each entry, its sign, and the support status. Status codes: SOLID = direct, often multiply-sourced or causally established primary support; SUPPORTED = a real primary citation exists, in some cases with a noted dose-, region-, or context-dependence caveat; EXCLUDED = a candidate term that was considered and dropped because the literature points the wrong way (see §6.2 for the alpha- and theta-closure cases). Every active entry below carries SOLID or SUPPORTED status; none rests on an unverified or contested citation.

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
Gamma	DA (phasic)	+	SUPPORTED (context-dependen dent)	Phasic VTA-dopamine stimulation increases PFC high-gamma power and theta–gamma coupling; D2 stimulation increases PFC gamma (Ott, Westendorff & Nieder, 2018, <i>J Cogn Neurosci</i>

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
				30:770–784; Lohani et al., 2019, <i>Cell Rep</i> 27(1):99–114). Caveat: direction is region/subtype- dependent (bath DA <i>decreases</i> gamma in motor cortex). Defensible in the PFC/encoding context, not as a universal law.
Gamma	GLU (phasic)	+	SOLID	Glutamatergic drive of PV-interneuron gamma is mechanistically established (Bartos, Vida, & Jonas, 2007; Buzsáki & Wang, 2012).
Gamma	ACh (phasic)	+	SUPPORTED	Cholinergic activation enhances gamma power — M1 muscarinic (Betterton et al., 2017, <i>Eur J</i> <i>Neurosci</i> , DOI 10.1111/ejn.13 582) and $\alpha 4\beta 2/\alpha 7$ nicotinic (Wang et al., 2015).

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
				Biphasic caveat: enhancement at low/physiologic al levels, suppression at high agonist concentrations.
Theta	ACh (total)	+	SOLID	Septal cholinergic input is the canonical theta generator (Buzsáki, 2002; Vertes & Kocsis, 1997); cholinergic tone positively tracks theta power in vivo (Zhang, Karadas, et al., 2024, <i>Neuron</i>). Replaces the excluded OXT/5-HT theta terms — see §6.2.
Theta	DA (tonic)	+	SOLID	Hippocampal– VTA loop and dopaminergic gating of theta (Lisman & Grace, 2005).
Alpha	GABA × NE (conjunctive)	+	SOLID	Alpha as the GABA-mediate d inhibitory rhythm of <i>relaxed</i> <i>wakefulness</i> ,

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
				gated by noradrenergic arousal (Klimesch, 2012; Bazanova & Vernon, 2014). The conjunctive form makes alpha a waking rhythm that collapses as LC/NE output falls, fixing the sleep/wake direction a tonic-GABA-only term gets backwards (§3.5.2, §6.2).
Alpha	GLU (phasic), desync	-	SUPPORTED	Active sensory/cognitive processing desynchronizes alpha; the phasic-glutamate term captures the falling limb of alpha's inverted-U arousal dependence (partial; §6.2).
Alpha	5-HT (tonic)	+	EXCLUDED	Dominant causal evidence points negative: 5-HT2A

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
				activation (psilocybin) <i>suppresses</i> posterior alpha power, reversed by the 5-HT2A antagonist ketanserin (Kometer et al., 2013, <i>J Neurosci</i> 33(25):10544; Muthukumaras wamy et al., 2013). Tonic/SSRI serotonergic effects are mixed. Excluded from the alpha closure — see §6.2.
Alpha	CB1 (tonic)	+	EXCLUDED	CB1's documented oscillatory effects are on theta and gamma, not alpha: a human CB1-genetics EEG study found <i>no</i> CB1 effect on alpha, and a rat CB1 agonist <i>decreased</i> alpha power (reversed by AM251).

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
Beta	NE (total)	+	SOLID	Excluded from the alpha closure — see §6.2. Beta as adaptive-gain/vigilance rhythm driven by locus-coeruleus NE (Aston-Jones & Cohen, 2005; Engel & Fries, 2010).
Beta	cortisol (total)	+	SUPPORTED	Corticosterone increases beta (and gamma) power, GR-mediated and causal — dexamethasone (GR agonist) mimics, mifepristone (GR antagonist) blocks; acute stress elevates CORT with positive correlation to oscillation power (Donaire et al., 2025, <i>Neuroscience</i> 585:144–157). Cortisol also enhances delta–beta coupling (Schutter & van

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
				Honk, 2005, <i>Int J Psychophysiol</i> 55(2):127–129). Inverted-U dose caveat.
Beta	ACh (total)	+	SUPPORTED	Activation of basal-forebrain cholinergic neurons increases beta (12–20 Hz) and low-gamma power (Wang et al., 2021, <i>Front Neurosci</i> 15:664410); cholinergic modulation of beta is linked to spatial attention and sensory processing.
Delta	MOR (tonic)	+	SUPPORTED (dose-dependent)	μ -opioid agonists at higher doses produce delta/slow-wave synchrony and sedation (high-voltage delta with morphine in the classic μ /k dissociation studies; opioids drive dissociated slow-wave/delta states,

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
				O'Brien et al., 2021, <i>J Neurophysiol</i> 126(4):1265–1275). Caveat: dose-dependent — low-dose μ is wake-promoting, and some high-dose slow-wave effect is kappa-mediated. Defensible in the high-opioid-tone NREM regime where MOR_tonic is used.
Delta	CB1 (tonic)	+	SUPPORTED	CB1 agonists increase NREM delta power, causally reversed by the CB1 antagonist AM251 (e.g., (m)VD-hemopressin pro-NREM study); consistent with CB1's documented pro-NREM role.
Delta	GABA (tonic)	+	SOLID	GABAergic slow-wave/delta generation (Steriade et al.,

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
Sigma	GABA (tonic)	+	SOLID	1993; Steriade, 2006). TRN GABAergic bursting underlies thalamocortical spindles (Steriade, 2006).
Sigma	GLU (phasic)	+	SOLID	Glutamatergic thalamocortical relay-cell rebound bursting is a core, well-characteri zed component of the spindle-generat ing loop (TRN GABA burst → TC glutamatergic rebound → cortical spindle; Steriade lineage; thalamocortical spindle reviews).
Sigma	NE (tonic)	-	SOLID (double-confir med)	Noradrenergic suppression of spindles confining them to NREM. Upgraded with direct optogenetic causal evidence: LC

Band	Transmitter (component)	Sign	Status	Candidate primary support / note
				silences permit spindle generation, and LC activation suppresses spindles by depolarizing thalamic/reticul ar circuits (Osorio-Forero et al., 2021, <i>Curr Biol</i> 31(22):5009–5 023; Swift et al., 2018, <i>Curr Biol</i> 28(22):3599–3 609; Steriade 2006 review). This negative sign is load-bearing for the no-sleep-flag property (§4.4.3); it now rests on causal, not just correlational, support.

Appendix D — Numerical methods

D.1 Boundary enforcement for the concentration components

The OU drift pushes each concentration toward its baseline , but the CIR-style diffusion term can in principle drive a discrete Euler–Maruyama step outside the admissible range . The implementation keeps both the tonic and phasic components in using the modular-folding reflecting scheme introduced in §3.2.5, rather than iterative clamping (which can diverge when the unconstrained step lands far outside the domain). Writing ,

for the bounds and for the domain width, a proposed post-step value is returned to the domain by

a single folding operation that reflects the trajectory off whichever boundary it crossed and remains correct for arbitrarily large excursions. The square-root diffusion is evaluated with a guard, , so the update never takes the square root of a transiently out-of-range pre-fold value. For the tonic component the Euler–Maruyama update before folding is

with the fold above then applied to ; the phasic component is handled identically.

In practice the fold is almost never triggered. The CIR/Feller condition for strict positivity of the continuous-time process is (Kloeden & Platen, 1992); under the default parameters of Appendix A, for all twelve systems, so the lower boundary at is reached with negligible probability, and the upper boundary at sits many stationary standard deviations above . The boundary scheme therefore governs worst-case robustness rather than the typical trajectory; folding, reflection, and hard clamping coincide whenever the boundary is not crossed.

D.2 Adaptive step-size via Richardson extrapolation

To control discretization error without committing to a single fixed , the framework supports adaptive step-sizing through Brownian-path Richardson extrapolation. At each proposed step of size :

1. Compute a *coarse* update: one Euler–Maruyama step of size using increment .
2. Compute a *fine* update: two Euler–Maruyama steps of size , using increments satisfying and . (The Brownian path is *split*, not independently resampled, so coarse and fine estimates are driven by the same noise realization.)
3. Compute the discrepancy .
4. If (configurable tolerance, default): reject the step, halve (to a minimum of), and retry from step 1.
5. If : accept (the higher-order estimate), and increase toward for the next step using a standard PI controller on .

The Brownian path splitting in step 2 is the key ingredient: it ensures the fine estimate is a genuine Richardson extrapolation of the coarse one, so the discrepancy is an honest local-error estimate rather than a noise artifact. Without splitting (i.e., independently sampling), would be dominated by the noise difference between the two estimates and the step-size controller would oscillate.

All integration in the framework uses `numpy.random.default_rng` with an explicit seed, so accepted steps are byte-reproducible regardless of the adaptive history: for the same seed and configuration, the trajectory is identical across runs.

D.3 Phasic burst: Brownian path construction

Phasic burst events add an instantaneous increment to the phasic component at the release time t . The burst amplitude A (eq. 14) is added in a single step that is not subject to the Richardson step-size test (the increment is deterministic at this point; the stochastic evolution begins on the subsequent step). The post-burst decay is then integrated under the adaptive scheme of §D.2, with the reflecting boundary of §D.1 applied throughout. This ensures that the phasic trajectory remains non-negative even when the burst amplitude is large relative to the local diffusion scale.

Appendix E — Per-receptor oscillatory coupling coefficients

This appendix tabulates the back-modulation direction of the closure loop (§3.6): how oscillatory band amplitudes feed back to modulate each receptor’s regulatory-state transition rates (the γ -accelerates-transitions coupling of eq. 41). Each receptor carries a list of `TransitionBandSpec(band,  $\lambda$ )` entries; the bands jointly scale that receptor’s CTMC transition rates through the multiplier

with the band amplitude. By the sign convention, a positive λ means elevation of that band correlates with faster state turnover / activity-dependent desensitization; a negative λ means it correlates with receptor stabilization. Coefficient magnitudes follow an evidence-graded heuristic: – for strong, mechanistically established, multiply-sourced associations; – for single-mechanism moderate associations; – for weak, indirect, or inferred couplings.

The coefficients below have been verified to match the implemented configuration exactly — all thirty receptors’ `transition_band_specs` agree between the source (`DEFAULT_RECEPTOR_CONFIGS`) and the literature-grounding table they were calibrated from. Every primary-source attribution below has been independently checked against the original publication. The Evidence column grades each entry: **P** = primary association (direct, individually verified empirical source); **Ind** = indirect (review-level, mechanism-level, or cross-band-extrapolated support rather than a band-specific causal primary); **Inf** = inferred from family/mechanism with no dedicated primary source.

#	Receptor	Band couplings (λ)	Primary source(s)	Ev.
1	DA_D1	$\gamma +0.4$, $\theta +0.3$, $\theta\gamma +0.4$	Stubbendorff et al. 2023; Ott, Westendorff & Nieder 2018	P
2	DA_D2	$\gamma +0.3$, $\theta +0.3$, $\beta +0.3$	Ott, Westendorff & Nieder 2018; Dejean et al. 2011	P

#	Receptor	Band couplings ()	Primary source(s)	Ev.
3	DA_D3	$\gamma -0.3, \theta -0.15$	Lemercier et al. 2016	P
4	DA_D4	$\gamma +0.15$	Andersson, Johnston & Fisahn 2012	P
5	DA_D5	$\gamma +0.3, \theta -0.2$	Albeely et al. 2023	P
6	5HT_1A	$\gamma -0.4, \theta -0.3, \beta -0.3, \delta +0.2$	Puig et al. 2010	P
7	5HT_1B	$\gamma -0.15, \theta -0.1$	inferred (5HT_1A family + autoreceptor)	Inf
8	5HT_2A	$\alpha -0.5, \beta -0.4, \theta -0.3, \delta -0.3, \gamma -0.3$	Kometer et al. 2013; Muthukumaraswamy et al. 2013	P
9	5HT_2C	$\alpha -0.3, \gamma -0.2$	Papp et al. 2020	P
10	5HT_3	$\gamma +0.3$	mechanistic (ionotropic fast-excitatory)	Inf
11	NE_alpha1	$\beta +0.4, \gamma +0.3$	Arnsten 2011	Ind
12	NE_alpha2	$\theta +0.3, \beta -0.2$	Wang et al. 2007	Ind
13	NE_beta1	$\beta +0.4, \gamma +0.3$	Arnsten 2011	Ind
14	NE_beta2	$\beta +0.2$	inferred (β -adrenergic family)	Inf
15	ACh_nic	$\gamma +0.4, \theta +0.4$	Wang et al. 2015; Lu & Henderson 2010	P
16	ACh_musc	$\theta +0.5, \gamma +0.3, \theta\gamma +0.5$	Betterton et al. 2017; Howe et al. 2017	P
17	OXTR	$\theta +0.4, \gamma +0.2$	Owen et al. 2013; Lin et al. 2018	Ind
18	MOR_mu	$\delta +0.3, \theta -0.2, \gamma -0.2$	McQuiston 2011	Ind
19	CB1	$\theta -0.3, \delta +0.3, \alpha\beta -0.2$	Hajós, Hoffmann & Kocsis 2008	P
20	CRH_R1	$\beta +0.4, \delta -0.3$	Romanowski et al. 2010; Steiger 2003	Ind
21	GABA_A	$\sigma +0.5, \alpha +0.4, \delta +0.3$	Uygun et al. 2022	P
22	GABA_B	$\delta +0.4, \theta -0.2$	Brown & McKenna 2015	Ind

#	Receptor	Band couplings ()	Primary source(s)	Ev.
2 3	GLU_NMD A	$\gamma +0.2$	Lemercier, Holman & Gerevich 2017	P
2 4	GLU_AMP A	$\gamma +0.5, \theta\gamma +0.4$	Buzsáki & Wang 2012	Ind
2 5	GLU_KAIN ATE	$\gamma +0.4$	Fisahn et al. 1998	P
2 6	GLU_mGlu R	$\theta +0.3, \gamma +0.2$	Whittington, Traub & Jefferys 1995	P
2 7	HIST_H1	$\gamma +0.4, \beta +0.3$	Andersson et al. 2017	P
2 8	HIST_H2	$\beta +0.3$	McCormick 1992 (review)	Ind
2 9	HIST_H3	$\theta -0.3$	Hajós et al. 2008	P
3 0	HIST_H4	$\gamma +0.1$	inferred (peripheral-dominant)	Inf

Legend: bands are δ (delta), θ (theta), α (alpha), β (beta), γ (gamma), σ (sigma), plus the composite envelopes $\theta\gamma$ (theta–gamma) and $\alpha\beta$ (alpha–beta). is the per-band coefficient in the transition-rate multiplier above.

Appendix F — Emotion-to-NT release-drive recipes

See Supplementary §S2 for the complete table of emotion-to-NT release-drive recipes.

Supplementary §S1 — Test-to-claim mapping

The framework ships with a suite of 1,440 neurochemistry-layer tests (the neurochem-specific subset of ZADOS v37's full test suite; ZADOS as a whole ships far more, covering unrelated cognitive engines). This section cross-references each core mathematical claim in §3 and §4 to the specific test(s) that verify it under seeded pseudorandom number generation. Per the framing of §6.3, these are implementation-fidelity tests, not biological-validity tests.

Paper claim	Section	Verifying test(s)	Test file
OU tonic component reverts to baseline under zero drive	§3.2 eq.(4)	test_compute_drift_term_at_baseline	test_mass_balance.py

Paper claim	Section	Verifying test(s)	Test file
Phasic concentration decays to zero between bursts	§3.2 eq.(5)	test_compute_mass_balance_drift_phasic_decay	test_mass_balance.py
CIR-style diffusion preserves $C(t) \geq 0$ (positivity)	§3.2 eq.(7)	test_diffusion_nonnegative	test_mass_balance.py
Burst amplitude saturates ($A_{burst} = A_{max}(1 - \exp(\dots))$)	§3.2 eq.(14)	test_compute_phasic_burst_amplitude_bounded	test_release_driver.py
Fatigue accumulates under repeated release	§3.2 eq.(15)	test_fatigue_accumulates	test_fatigue.py
Adaptive threshold rises with recent-activity trace	§4.2 eq.(46)	test_compute_adaptive_threshold_high_activity	test_release_driver.py
Michaelis–Menten saturation function	§3.3 eq.(20)	test_compute_saturation_half_saturation	test_receptor_dynamics.py
Effective-signaling proxy is product of 5 factors	§3.3 eq.(22)	test_gamma_in_full_formula	test_aij_gamma.py
ACTIVE $\rightarrow$ DESENSITIZED transition	§3.4	test_active_to_desensitized_eligible	test_receptor_dynamics.py
DESENSITIZED $\rightarrow$ INTERNALIZED transition	§3.4	test_desensitized_to_internalized_eligible	test_receptor_dynamics.py
INTERNALIZED $\rightarrow$ ACTIVE recycling	§3.4	test_internalized_to_active_recycling	test_receptor_dynamics.py
ACTIVE $\rightarrow$ UPREGULATED transition	§3.4	test_active_to_upregulated_eligible	test_receptor_dynamics.py
DESENSITIZED $\rightarrow$ ACTIVE recovery	§3.4	test_desensitized_to_active_recovery_eligible	test_receptor_dynamics.py
All 30 receptors carry band specs	Appendix B	test_every_receptor_has_transition_band_specs	test_pr5_transition_specs_cfc.py

Paper claim	Section	Verifying test(s)	Test file
Beta accelerates receptor state transitions	§3.5 eq.(41)	test_beta_accelerates_desensitization_in_engine	test_oscillation_coupling.py
Band closure bounded to [0,1]	§3.5 eq.(32)	test_clamped_to_01	test_oscillation_generator.py
CFC envelopes are products of constituent band amplitudes	§3.5 eq.(33–35)	test_cross_frequency_coupling	test_oscillation_generator.py
Steady-state empirical mean/std over 10^4 seeded runs (seed 20240615)	§4.1	(none)	figure-reproduction only, not pytest
Burst amplitude decreases with fatigue (mechanism; not the exact Fig. 4.2.2 sequence)	§4.2	test_apply_fatigue_gating_above_threshold	test_release_drives.py
Burst decay time prolonged by fatigue	§4.2	(none)	no dedicated test found; figure-reproduction only
Sustained $S > 0.7$ drive for $t_{\text{state}} > 5$ steps triggers desensitization	§4.3	test_active_to_desensitized_eligible, test_active_to_desensitized_not_eligible_insufficient_time	test_receptor_dynamics.py
Encoding regime is theta–gamma dominant	§4.5	test_encoding_regime_theta_gamma_dominant	test_oscillation_generator.py
Inhibitory-control regime is alpha–beta dominant	§4.5	test_inhibitory_regime_alpha_beta_dominant	test_oscillation_generator.py
NREM regime is delta–sigma dominant (delta confirmed; sigma	§4.5	test_nrem_alpha_collapses	test_oscillation_generator.py

Paper claim	Section	Verifying test(s)	Test file
not independently asserted by this test)			

Supplementary §52 — Emotion-to-NT release-drive recipes

The framework’s release-drive signal (§3.2.3) receives contributions from a weighted sum of cognitive signals including an emotion-derived component. When an upstream emotion profile is active, each emotion maps to a per-NT modulation via the DEFAULT_EMOTION_RECIPES table below. Each recipe specifies a dict of {signal_key: base_weight} per NT; the final release-drive contribution is the emotion strength times the base weight, summed across all active emotions. Negative weights suppress release (e.g., anxiety reduces GABA tone; sadness reduces DA and 5-HT). The twelve implemented structural emotions are the core set; the framework supports arbitrary custom recipes via the same EmotionNTRecipe interface.

Emotion	NT modulated	Signal key	Base weight	Notes
joy	DA	emotion_drive	+0.8	Positive affect burst
	5-HT	emotion_drive	+0.6	Stability component
	MOR	emotion_drive	+0.5	Opioid hedonia
	OXT	emotion_drive	+0.3	Social warmth
curiosity	DA	emotion_drive	+0.7	Novelty-seeking
	DA	novelty	+0.5	Novelty signal weight
	5-HT	emotion_drive	+0.4	5-HT2A symbolic expansion
	ACh	emotion_drive	+0.5	Attention engagement
	CB1	emotion_drive	+0.4	Cognitive flexibility

Emotion	NT modulated	Signal key	Base weight	Notes
anxiety	GLU	emotion_drive	+0.4	NMDA high-complexity binding
	Histamine	emotion_drive	+0.4	Arousal
	NE	emotion_drive	+0.7	Threat arousal
	CRH	emotion_drive	+0.6	Stress axis
	Cortisol	emotion_drive	+0.5	HPA activation
	DA	emotion_drive	+0.3	D2 inhibitory control
calm	GABA	emotion_drive	-0.3	Reduced inhibitory tone
	5-HT	emotion_drive	+0.7	Tonic stability
	GABA	emotion_drive	+0.6	Inhibitory tone
	MOR	emotion_drive	+0.4	Opioid comfort
empathy	NE	emotion_drive	-0.3	Reduced arousal
	OXT	emotion_drive	+0.8	Social bonding
	OXT	empathy	+0.5	Empathy signal weight
	5-HT	emotion_drive	+0.4	Mood
focus	MOR	emotion_drive	+0.3	Affiliative safety
	ACh	emotion_drive	+0.8	Attentional precision
	ACh	attention_demand	+0.5	Demand weight

Emotion	NT modulated	Signal key	Base weight	Notes
sadness	NE	emotion_drive	+0.5	Arousal
	NE	precision	+0.3	Precision weight
	DA	emotion_drive	+0.3	Motivation
	GABA	emotion_drive	+0.3	Irrelevant-input suppression
	Histamine	emotion_drive	+0.6	Wakefulness
	Histamine	wakefulness	+0.4	Wakefulness weight
	5-HT	emotion_drive	-0.4	Reduced serotonin tone
	DA	emotion_drive	-0.3	Anhedonia
anger	MOR	emotion_drive	+0.4	Opioid buffering
	OXT	emotion_drive	+0.3	Social seeking
	NE	emotion_drive	+0.8	Approach aggression
	DA	emotion_drive	+0.5	Drive
	CRH	emotion_drive	+0.4	Stress activation
trust	GABA	emotion_drive	-0.4	Disinhibition
	OXT	emotion_drive	+0.8	Bonding
	OXT	trust	+0.6	Trust signal weight
	5-HT	emotion_drive	+0.5	Stability
	MOR	emotion_drive	+0.3	Affiliative safety

Emotion	NT modulated	Signal key	Base weight	Notes
surprise	DA	emotion_drive	+0.3	Exploratory openness
	GABA	emotion_drive	+0.3	Calm
	NE	emotion_drive	+0.7	Salience / orienting
	DA	emotion_drive	+0.5	Reward prediction error
	DA	rpe	+0.3	RPE weight
	ACh	emotion_drive	+0.4	Attention
	GLU	emotion_drive	+0.3	Integration
contentment	Histamine	emotion_drive	+0.5	Alertness
	5-HT	emotion_drive	+0.7	Tonic stability
	MOR	emotion_drive	+0.6	Hedonia
	GABA	emotion_drive	+0.4	Calm
fear	DA	emotion_drive	+0.2	Mild positive affect
	NE	emotion_drive	+0.9	Maximal threat arousal
	CRH	emotion_drive	+0.8	HPA — acute stress
	Cortisol	emotion_drive	+0.6	Cortisol mobilization
	GABA	emotion_drive	-0.5	Disinhibition under threat
	DA	emotion_drive	-0.2	Suppressed exploratory DA

Source: DEFAULT_EMOTION_RECIPES dict, zados/neurochem/emotion_interface.py (ZADOS v36). All values extracted directly from the implementation; none have been modified. Cortisol in anxiety and fear reflects the CRH–cortisol pathway — CRH drives cortisol release; both are represented as independent NT drives in the framework. Per §1, these recipes are engineering choices about how affective states map to neurochemical responses, not empirical fits; they are consistent with the qualitative pharmacology literature but their specific magnitudes are calibration points open to revision.